

Spring 2011

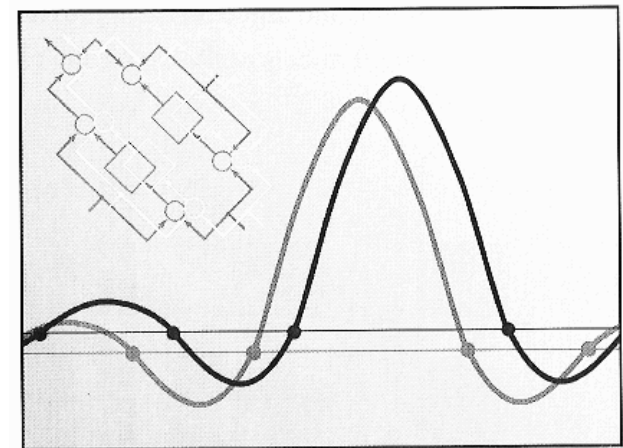
信號與系統 Signals and Systems

Chapter SS-4 The Continuous-Time Fourier Transform

Feng-Li Lian

NTU-EE

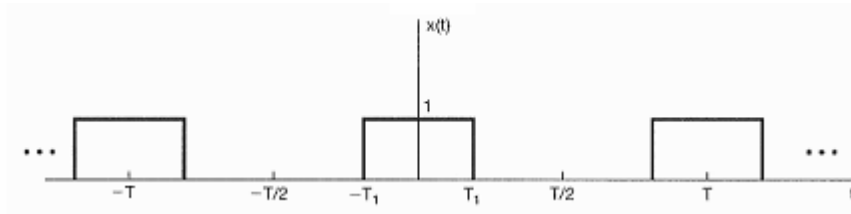
Feb11 – Jun11



Figures and images used in these lecture notes are adopted from
"Signals & Systems" by Alan V. Oppenheim and Alan S. Willsky, 1997

- Representation of **Aperiodic** Signals:
the Continuous-Time Fourier Transform
- The Fourier Transform for **Periodic** Signals
- **Properties**
of the Continuous-Time Fourier Transform
- The **Convolution** Property
- The **Multiplication** Property
- **Systems** Characterized by
Linear Constant-Coefficient Differential Equations

■ Example 3.5: $a_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_T x(t) e^{-jk(2\pi/T)t} dt$



$$x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & T_1 < |t| < T/2 \end{cases}$$

$$k = 0 \quad a_0 = \frac{1}{T} \int_{-T_1}^{T_1} dt = \frac{2T_1}{T}$$

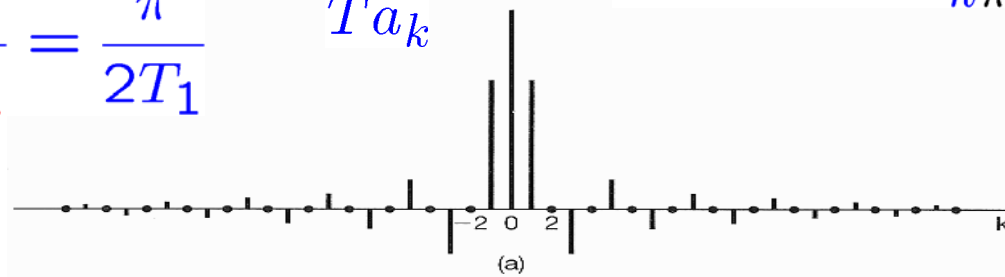
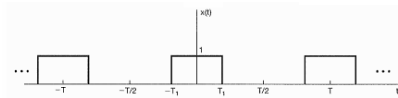
$$k \neq 0 \quad a_k = \frac{1}{T} \int_{-T_1}^{T_1} e^{-jk\omega_0 t} dt = \frac{1}{T} \frac{1}{(-jk\omega_0)} e^{-jk\omega_0 t} \Big|_{-T_1}^{T_1}$$

$$= \frac{1}{jk\omega_0 T} [e^{jk\omega_0 T_1} - e^{-jk\omega_0 T_1}] / \quad \omega_0 = \frac{2\pi}{T}$$

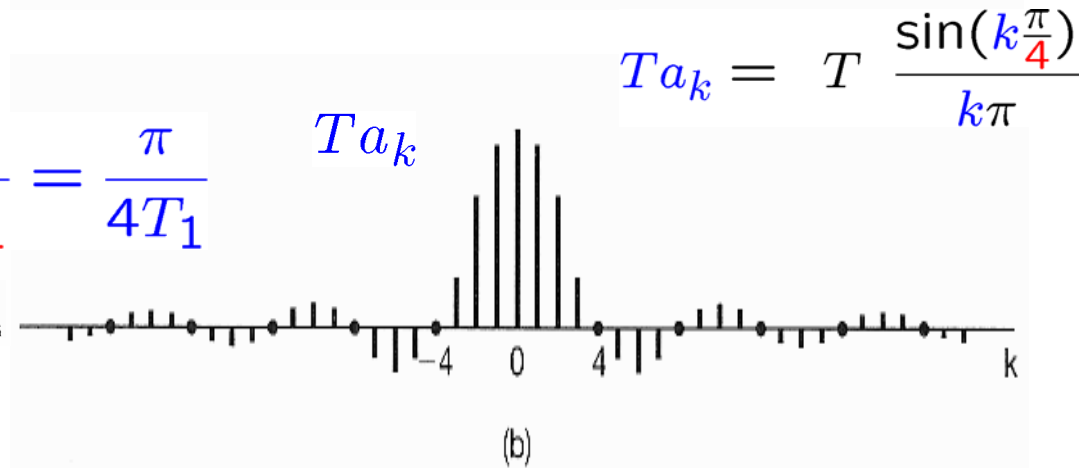
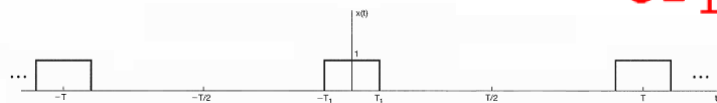
$$= \frac{2 \sin(k\omega_0 T_1)}{k\omega_0 T} = \frac{\sin(k\omega_0 T_1)}{k\pi} = \frac{\sin(k(2\pi/T)T_1)}{k\pi}, \quad k \neq 0$$

■ Example 3.5: $T a_k = T \frac{\sin(k 2\pi \frac{T_1}{T})}{k\pi}$ $T a_k = T \frac{\sin(k \frac{\pi}{2})}{k\pi}$

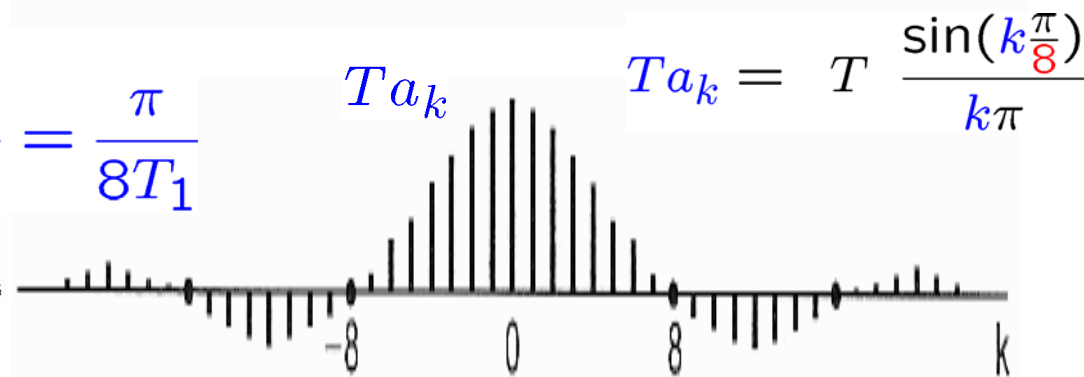
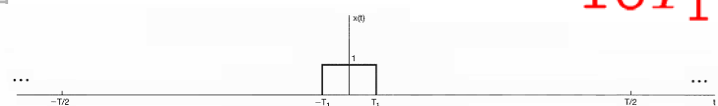
$T = 4T_1$ $w_0 = \frac{2\pi}{4T_1} = \frac{\pi}{2T_1}$



$T = 8T_1$ $w_0 = \frac{2\pi}{8T_1} = \frac{\pi}{4T_1}$



$T = 16T_1$ $w_0 = \frac{2\pi}{16T_1} = \frac{\pi}{8T_1}$

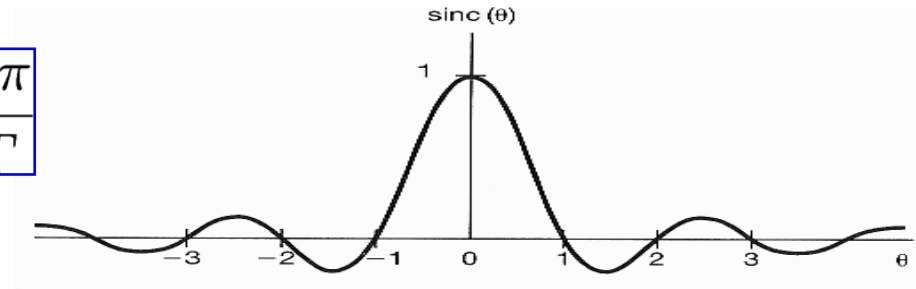


■ Example 3.5:

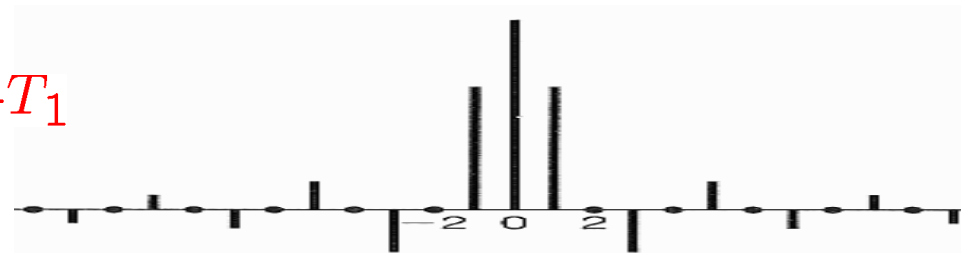
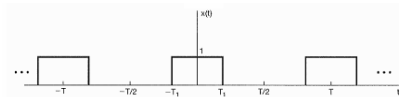
$$Ta_k = T \frac{2 \sin(k\omega_0 T_1)}{k\omega_0 T}$$

$$= T_1 \frac{2 \sin(k\omega_0 T_1)}{k\omega_0 T_1}$$

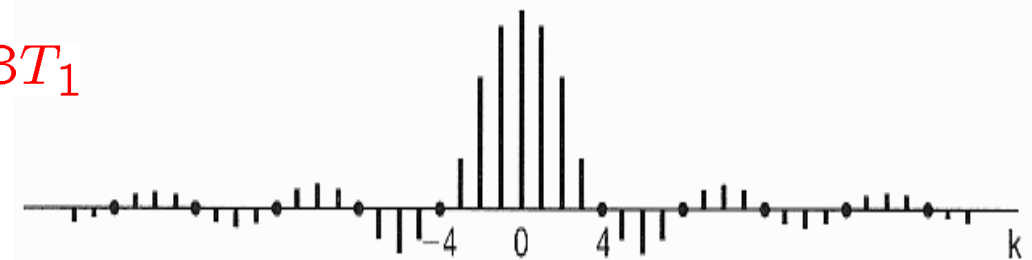
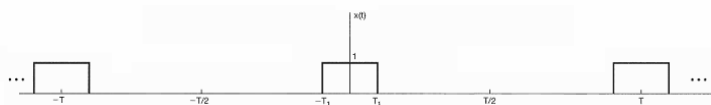
$$\omega_0 = \frac{2\pi}{T}$$



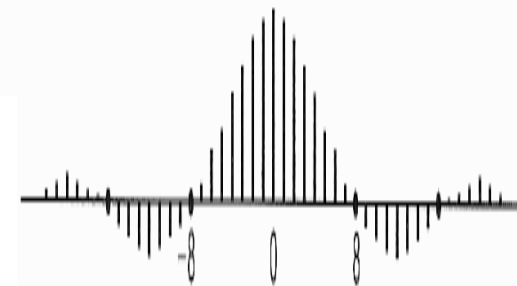
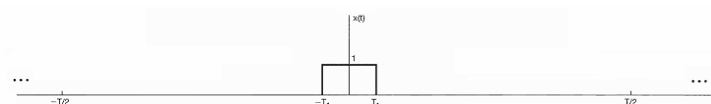
$$T = 4T_1$$



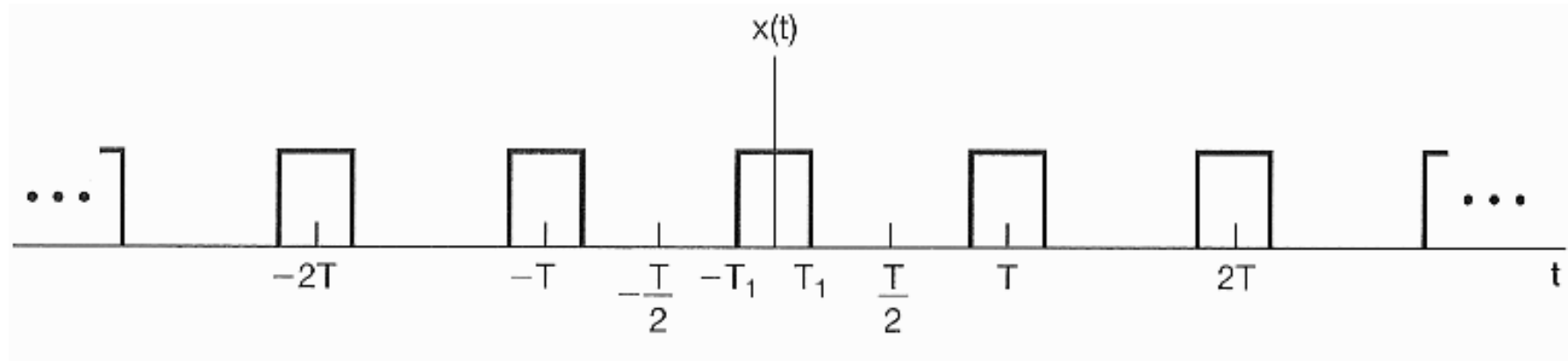
$$T = 8T_1$$



$$T = 16T_1$$



CT Fourier Transform of an Aperiodic Signal:



$$x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & T_1 < |t| < T/2 \end{cases}$$

$$a_k = \frac{2 \sin(k\omega_0 T_1)}{k\omega_0 T}$$

Fourier series coefficients

$$T a_k = \left. \frac{2 \sin(\omega T_1)}{\omega} \right|_{\omega = k\omega_0}$$

ω as a continuous variable

Representation of Aperiodic Signals: CT Fourier Transform

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$$Ta_k = \frac{2 \sin(\omega T_1)}{\omega}$$

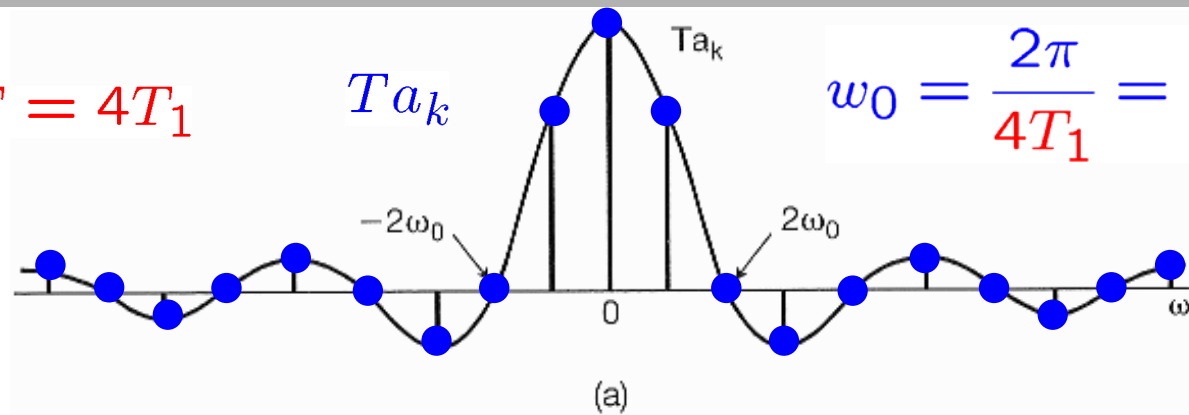
$$\omega = k\omega_0 = k \frac{2\pi}{T}$$

$$\omega_0 = \frac{2\pi}{T}$$

$$T = 4T_1$$

Ta_k

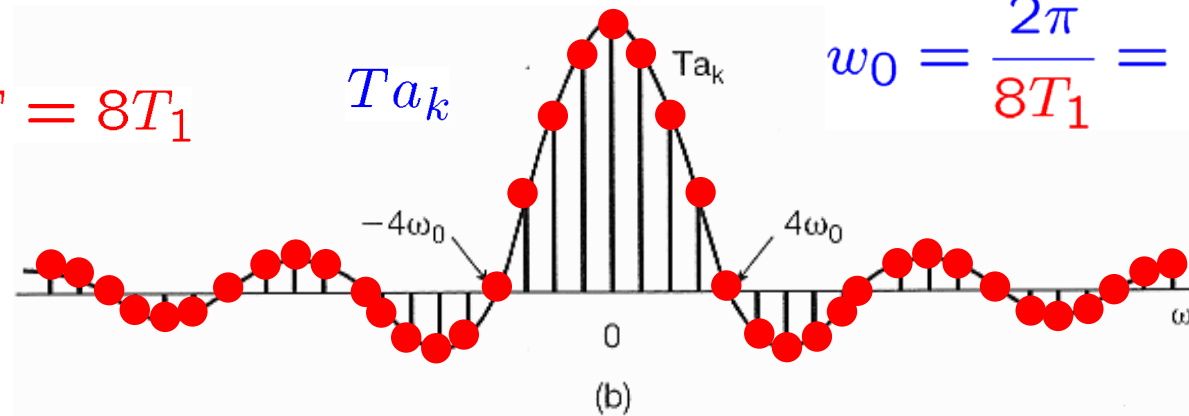
$$\omega_0 = \frac{2\pi}{4T_1} = \frac{\pi}{2T_1}$$



$$T = 8T_1$$

Ta_k

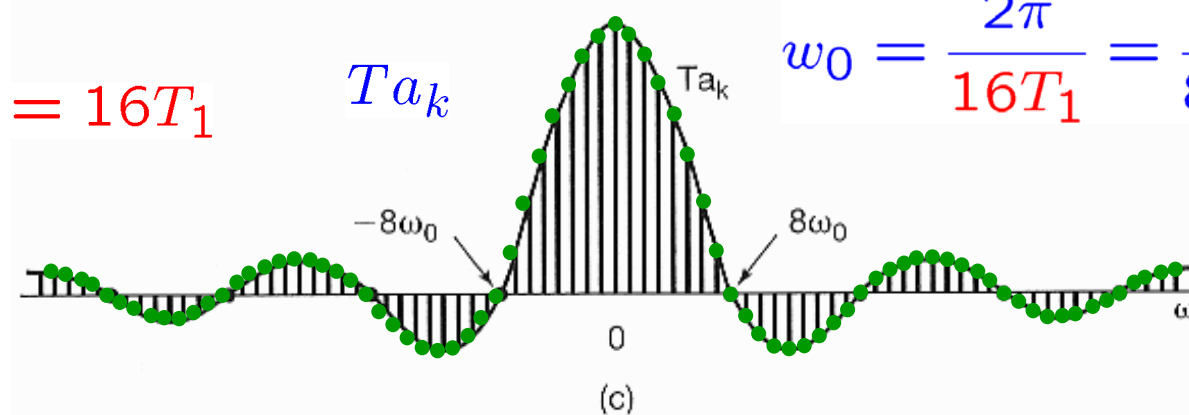
$$\omega_0 = \frac{2\pi}{8T_1} = \frac{\pi}{4T_1}$$



$$T = 16T_1$$

Ta_k

$$\omega_0 = \frac{2\pi}{16T_1} = \frac{\pi}{8T_1}$$

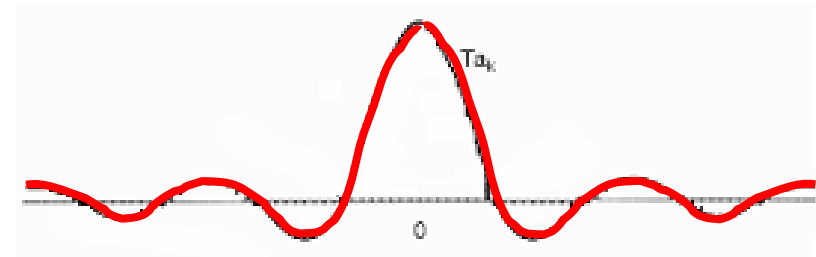
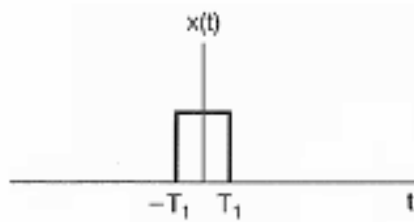
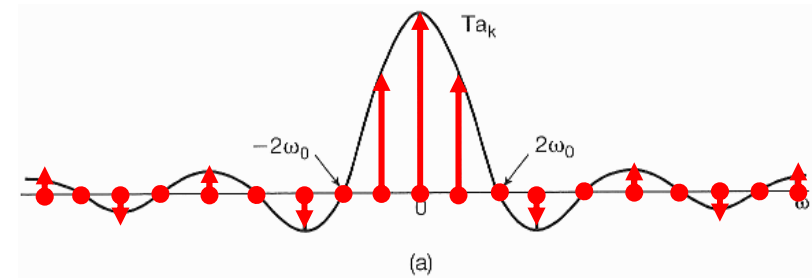
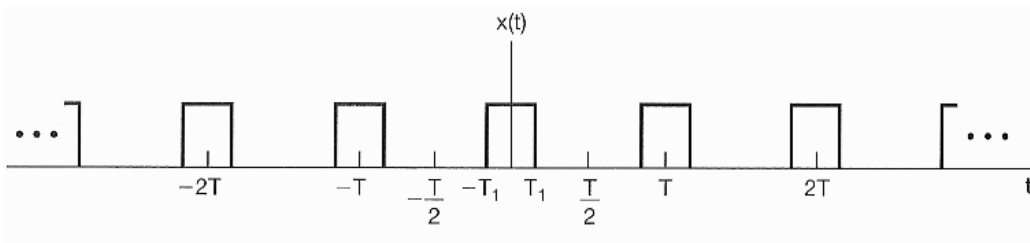


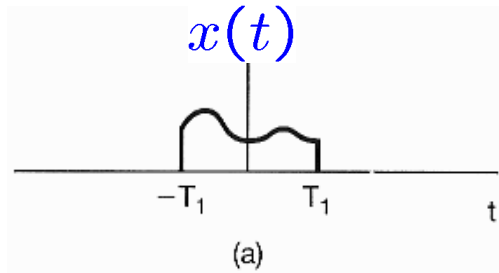
Representation of Aperiodic Signals: CT Fourier Transform

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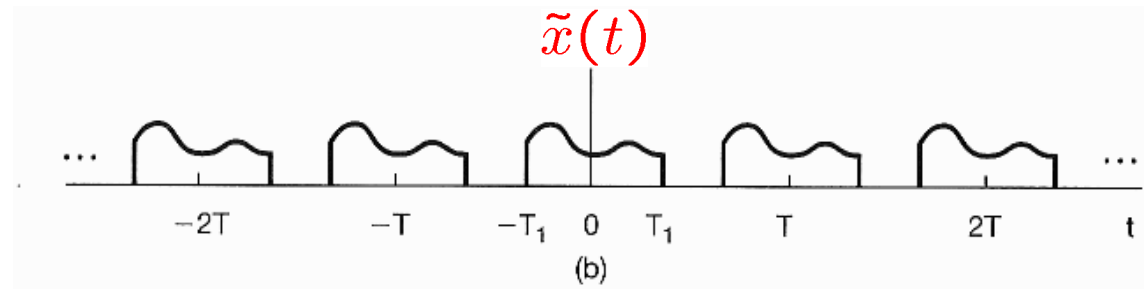
$$\omega = k\omega_0 = k\frac{2\pi}{T}$$

$$T \rightarrow \infty \Rightarrow \{Ta_k\} \rightarrow \left. \frac{2 \sin(\omega T_1)}{\omega} \right|_{\omega = k\omega_0}$$





an aperiodic signal



a periodic signal

$$\tilde{x}(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$$

$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} \tilde{x}(t) e^{-jk\omega_0 t} dt$$

$$\Rightarrow a_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_{-\infty}^{+\infty} x(t) e^{-jk\omega_0 t} dt$$

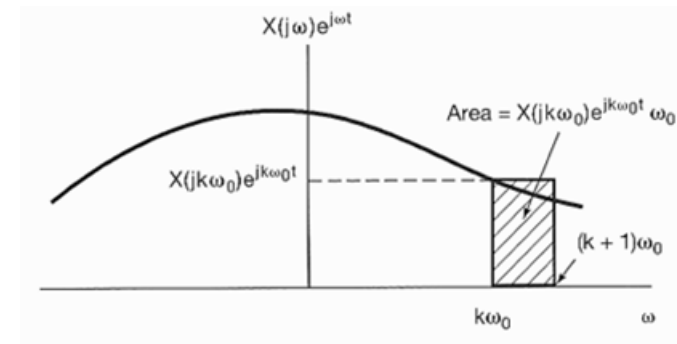
- Define the envelope $X(j\omega)$ of Ta_k as

$$Ta_k = \frac{2 \sin(\omega T_1)}{\omega}$$

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$$

- Then,

$$a_k = \frac{1}{T} X(jk\omega_0)$$



- Hence,

$$\tilde{x}(t) = \sum_{k=-\infty}^{+\infty} \frac{1}{T} X(jk\omega_0) e^{jk\omega_0 t}$$

$$\omega_0 = \frac{2\pi}{T}$$

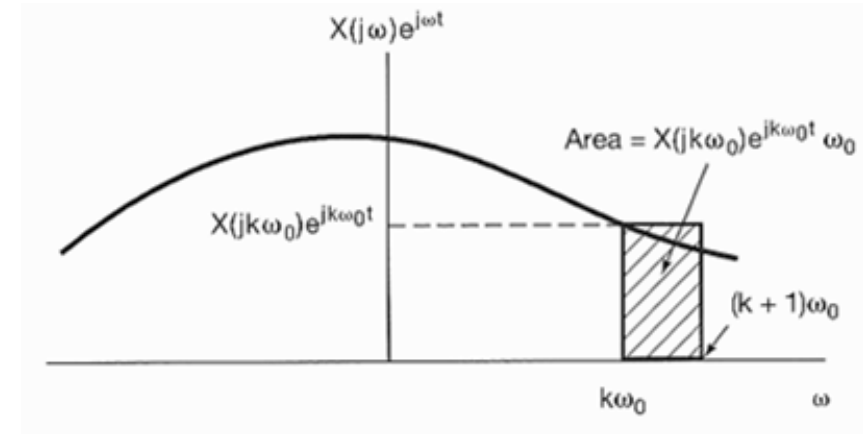
$$\frac{1}{T} = \frac{1}{2\pi} \omega_0$$

$$= \frac{1}{2\pi} \sum_{k=-\infty}^{+\infty} X(jk\omega_0) e^{jk\omega_0 t} \omega_0$$

- As $T \rightarrow \infty$, $\tilde{x}(t) \rightarrow x(t)$

also $\omega_0 \rightarrow 0$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$$



- inverse Fourier transform eqn

- synthesis eqn

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$$

- $X(j\omega)$: Fourier Transform of $x(t)$
spectrum

- analysis eqn

$$a_k = \frac{1}{T} X(j\omega) \Big|_{\omega=k\omega_0}$$

■ Sufficient conditions for the convergence of FT

$$x(t) \xrightarrow{CT\mathcal{F}T} X(j\omega)$$

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$$

$$\hat{x}(t) \xleftarrow{CT\mathcal{I}FT} X(j\omega)$$

$$\hat{x}(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$$

$$e(t) = \hat{x}(t) - x(t)$$

- If $x(t)$ has finite energy

i.e., square integrable, $\int_{-\infty}^{+\infty} |x(t)|^2 dt < \infty$

$$\Rightarrow X(j\omega) \text{ is finite}$$

$$\Rightarrow \int_{-\infty}^{+\infty} |e(t)|^2 dt = 0$$

■ Sufficient conditions for the convergence of FT

- Dirichlet conditions:

1. $x(t)$ be absolutely integrable; that is, $\int_{-\infty}^{+\infty} |x(t)| dt < \infty$

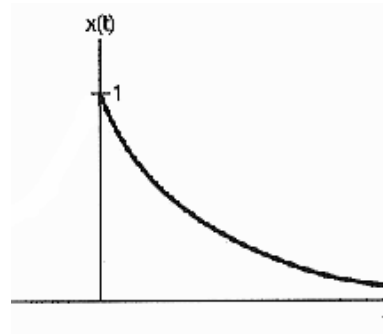
2. $x(t)$ have a finite number of maxima and minima within any finite interval

3. $x(t)$ have a finite number of discontinuities within any finite interval

Furthermore, each of these discontinuities must be finite

■ Example 4.1:

$$x(t) = e^{-at} u(t), \quad a > 0$$



$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$\Rightarrow X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} e^{-at} u(t) e^{-j\omega t} dt$$

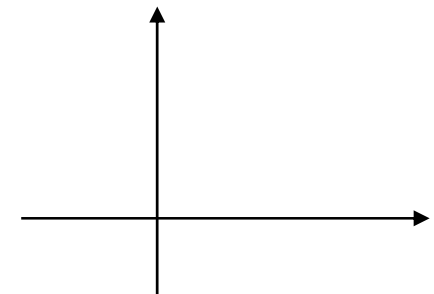
$$= \int_0^{\infty} e^{-at} e^{-j\omega t} dt$$

$$= \int_0^{\infty} e^{-(a+j\omega)t} dt$$

$$= -\frac{1}{a + j\omega} e^{-(a+j\omega)t} \Big|_0^{\infty}$$

$$= 0 - \left(-\frac{1}{a + j\omega} e^{-(a+j\omega)0} \right)$$

$$= \frac{1}{a + j\omega}, \quad a > 0$$

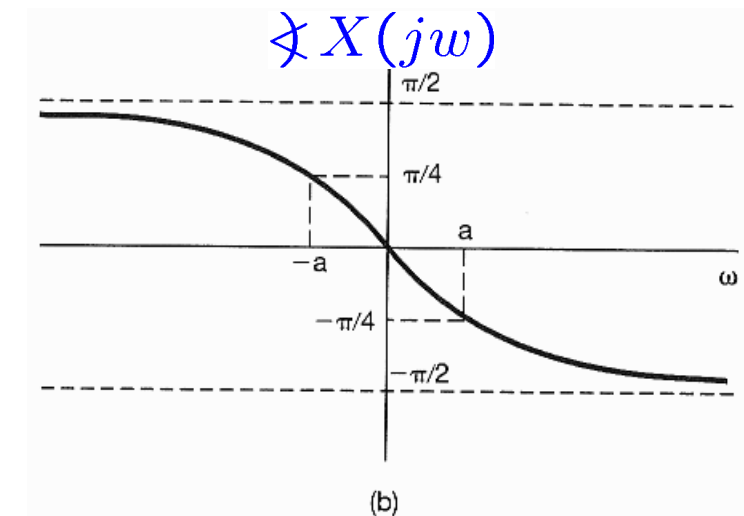
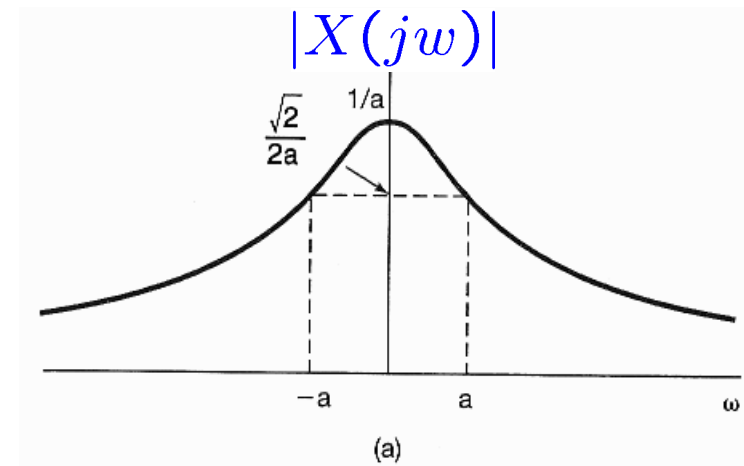


■ Example 4.1:

$$\Rightarrow X(j\omega) = \frac{1}{a + j\omega}, \quad a > 0$$

$$\Rightarrow |X(j\omega)| = \frac{1}{\sqrt{a^2 + \omega^2}}$$

$$\Rightarrow \angle X(j\omega) = -\tan^{-1} \left(\frac{\omega}{a} \right)$$



■ Example 4.2:

$$x(t) = e^{-a|t|}, \quad a > 0$$

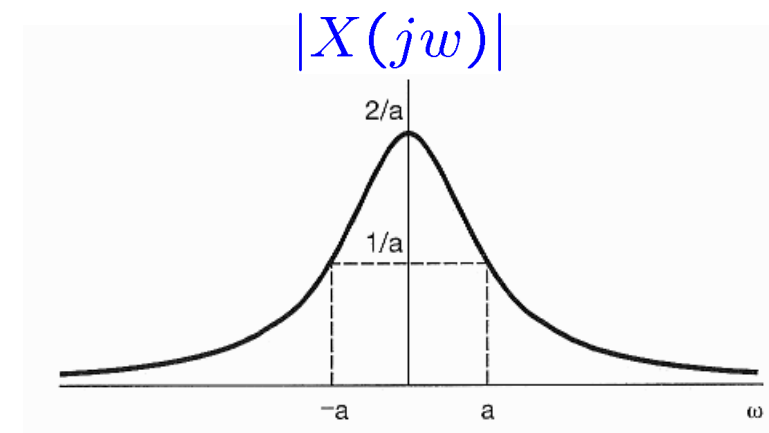
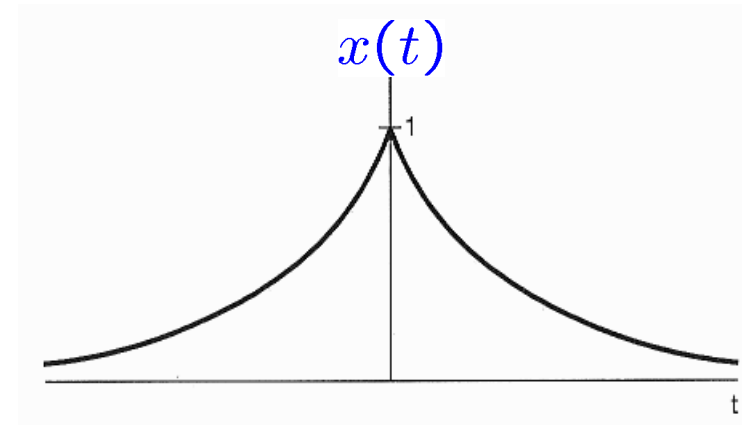
$$\Rightarrow X(j\omega) = \int_{-\infty}^{\infty} e^{-a|t|} e^{-j\omega t} dt$$

$$= \int_{-\infty}^0 e^{at} e^{-j\omega t} dt + \int_0^{\infty} e^{-at} e^{-j\omega t} dt$$

$$= \frac{1}{a - j\omega} + \frac{1}{a + j\omega}$$

$$= \frac{2a}{a^2 + \omega^2}$$

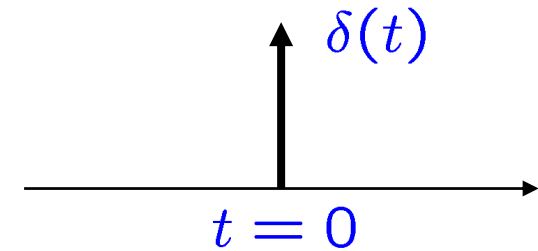
$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$



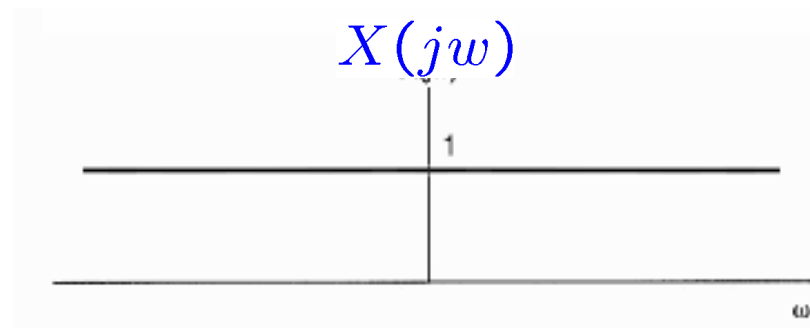
$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

■ Example 4.3:

$x(t) = \delta(t)$, i.e., unit impulses



$$\Rightarrow X(j\omega) = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = 1$$



■ Example 4.4:

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & |t| > T_1 \end{cases}$$

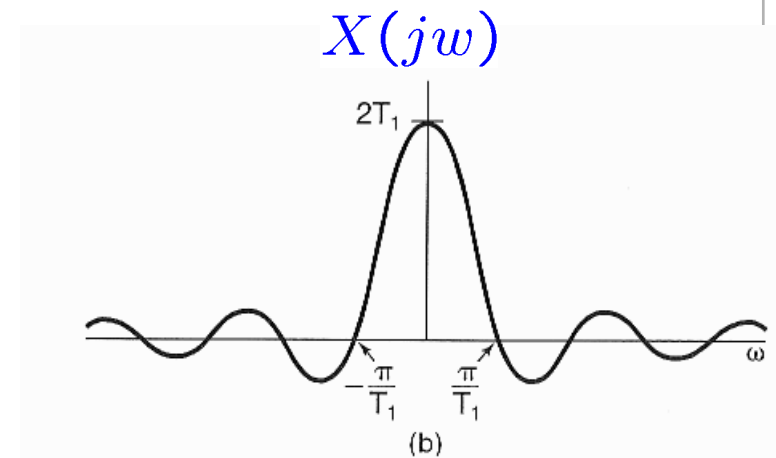
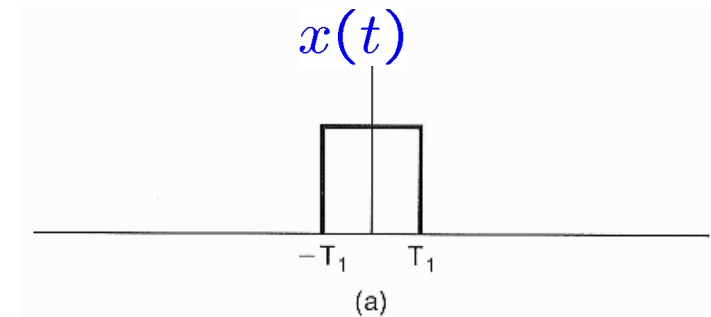
$$\Rightarrow X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = \int_{-T_1}^{T_1} e^{-j\omega t} dt$$

$$= \frac{1}{-j\omega} e^{-j\omega t} \Big|_{-T_1}^{T_1}$$

$$= \frac{1}{-j\omega} \left(e^{-j\omega T_1} - e^{j\omega T_1} \right)$$

$$= \frac{1}{j\omega} \left(e^{j\omega T_1} - e^{-j\omega T_1} \right)$$

$$= 2 \frac{\sin(\omega T_1)}{\omega} = 2T_1 \frac{\sin(\pi \omega T_1 / \pi)}{\pi \omega T_1 / \pi}$$



■ Example 4.5:

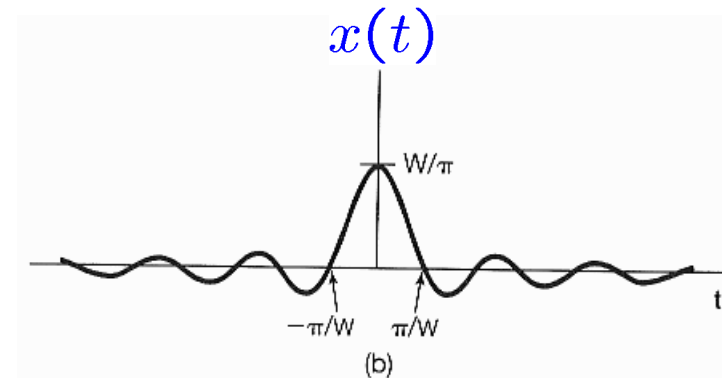
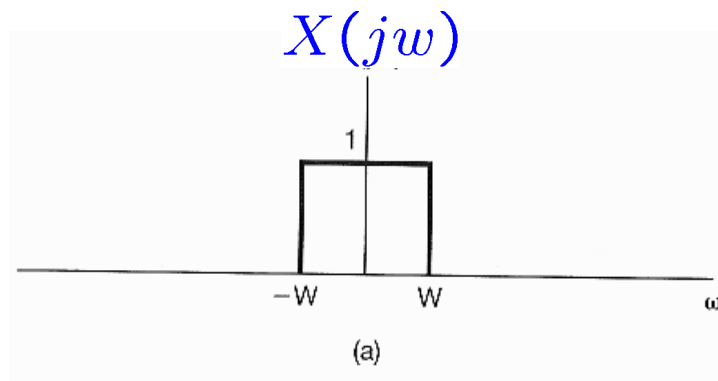
$$X(j\omega) = \begin{cases} 1, & |\omega| < W \\ 0, & |\omega| > W \end{cases}$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$\Rightarrow x(t) = \frac{1}{2\pi} \int_{-W}^W e^{j\omega t} d\omega$$

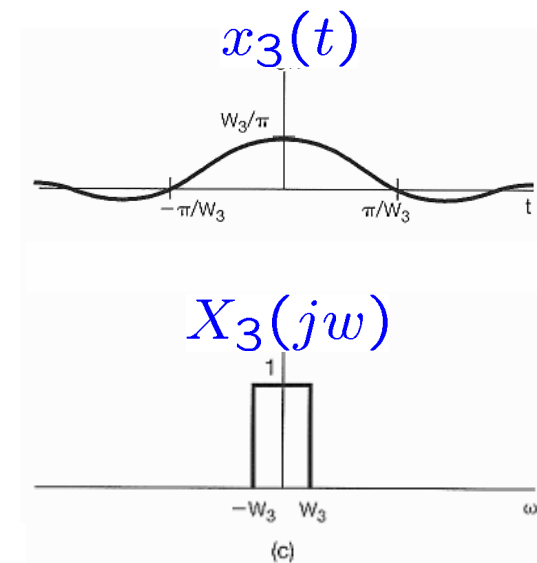
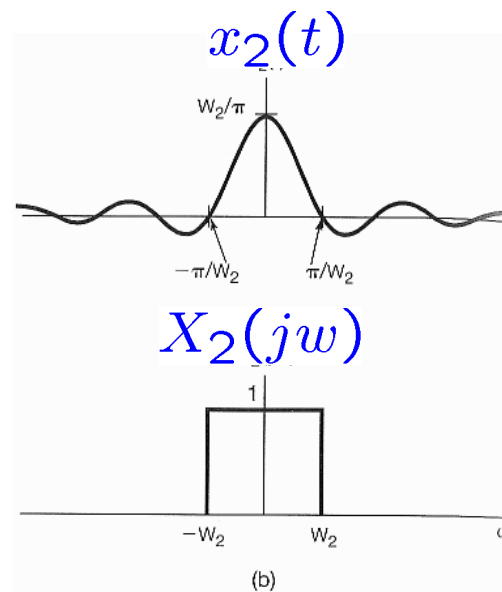
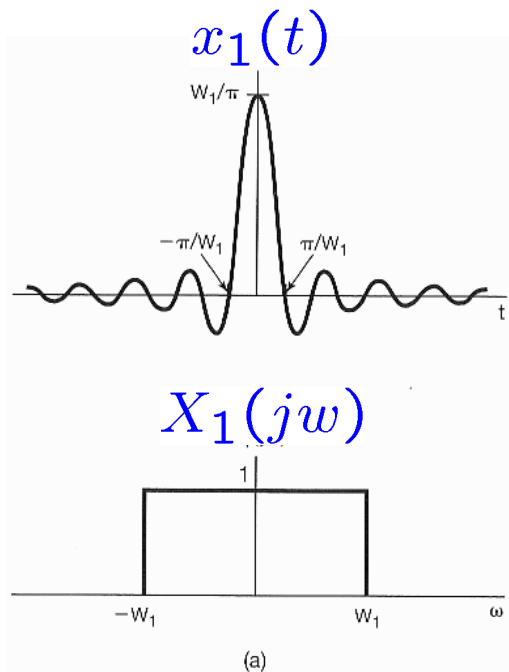
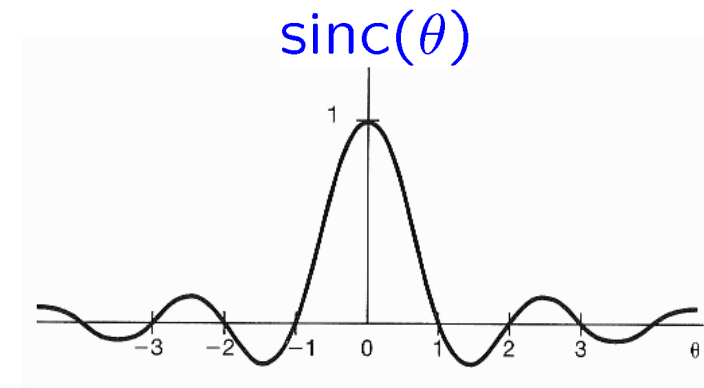
$$= \frac{\sin(Wt)}{\pi t}$$



■ sinc functions:

$$\text{sinc}(\theta) = \frac{\sin(\pi\theta)}{\pi\theta}$$

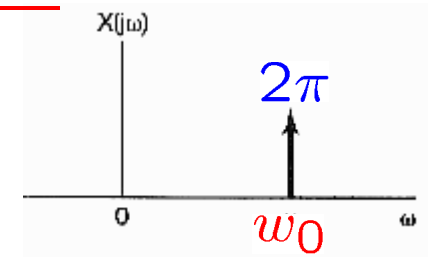
$$\frac{\sin(Wt)}{\pi t} = \frac{W}{\pi} \text{sinc}\left(\frac{Wt}{\pi}\right)$$



- Representation of **Aperiodic** Signals:
the Continuous-Time Fourier Transform
- The Fourier Transform for **Periodic** Signals
- Properties
of the Continuous-Time Fourier Transform
- The Convolution Property
- The Multiplication Property
- Systems Characterized by
Linear Constant-Coefficient Differential Equations

■ Fourier Transform from Fourier Series:

$$X(j\omega) = 2\pi \delta(\omega - \omega_0)$$

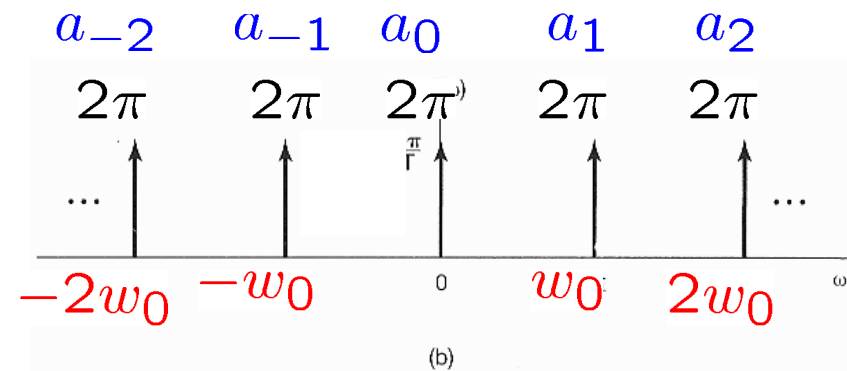


$$\Rightarrow x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} 2\pi \delta(\omega - \omega_0) e^{j\omega t} d\omega$$

$$= e^{j\omega_0 t}$$

- more generally,

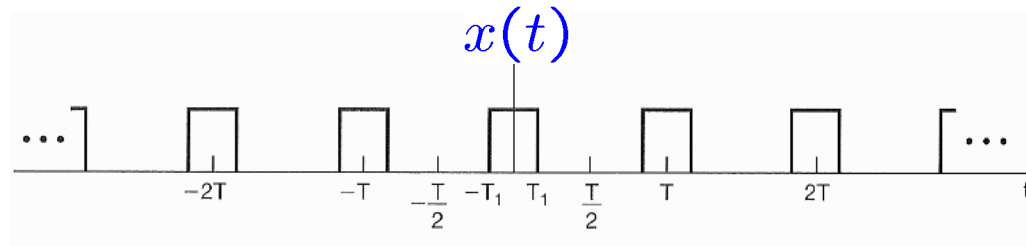
$$X(j\omega) = \sum_{k=-\infty}^{+\infty} 2\pi a_k \delta(\omega - k\omega_0)$$



$$\Rightarrow x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$$

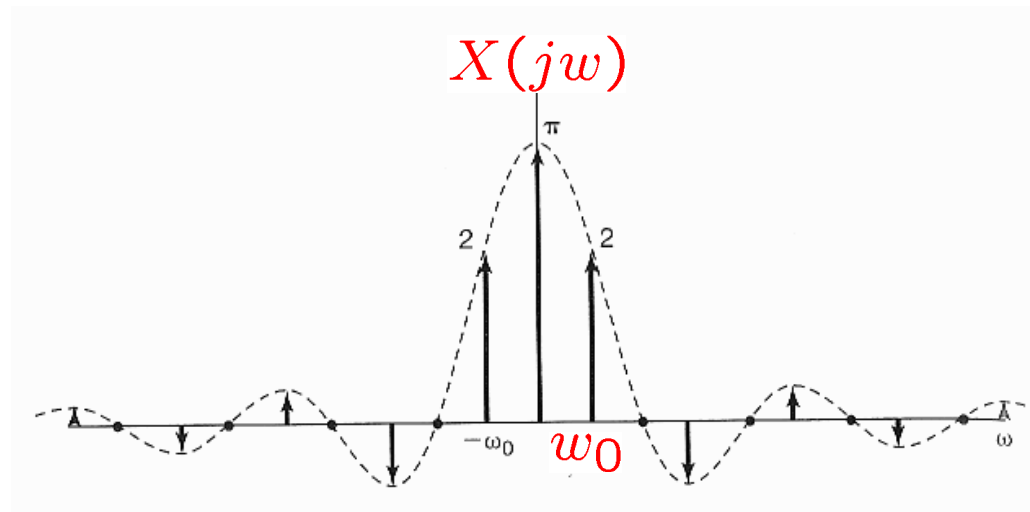
Fourier series representation
of a periodic signal

■ Example 4.6:



$$\Rightarrow a_k = \frac{\sin(k\omega_0 T_1)}{\pi k}$$

$$\Rightarrow X(j\omega) = \sum_{k=-\infty}^{+\infty} 2\pi a_k \delta(\omega - k\omega_0) = \sum_{k=-\infty}^{+\infty} \frac{2 \sin(k\omega_0 T_1)}{k} \delta(\omega - k\omega_0)$$



■ Example 4.7:

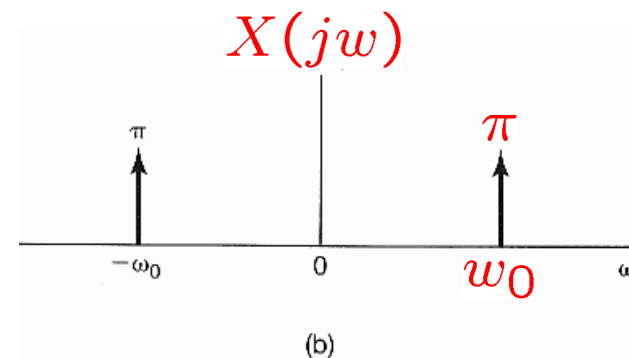
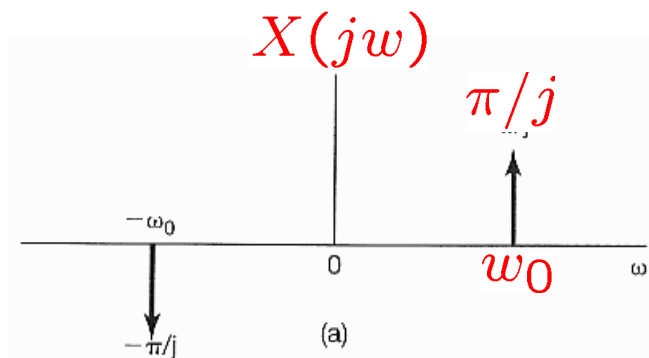
$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$$

$$x(t) = \sin(\omega_0 t) = \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j}$$

$$\Rightarrow a_1 = \frac{1}{2j} \quad a_{-1} = -\frac{1}{2j} \quad a_k = 0, \quad k \neq 1, -1$$

$$x(t) = \cos(\omega_0 t) = \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2}$$

$$\Rightarrow a_1 = \frac{1}{2} \quad a_{-1} = \frac{1}{2} \quad a_k = 0, \quad k \neq 1, -1$$



■ Example 4.8:

$$x(t) = \sum_{k=-\infty}^{+\infty} \delta(t - kT)$$

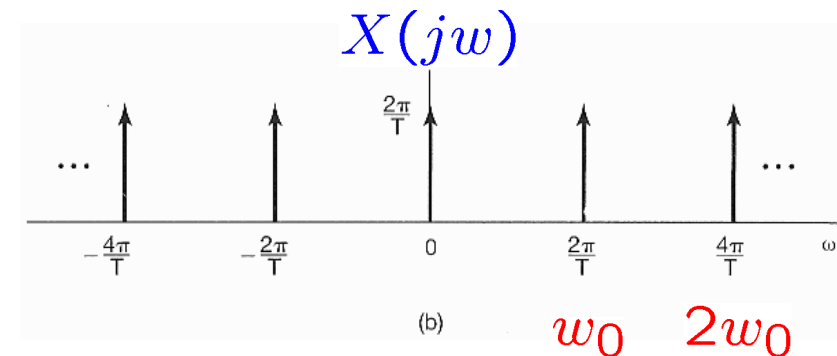
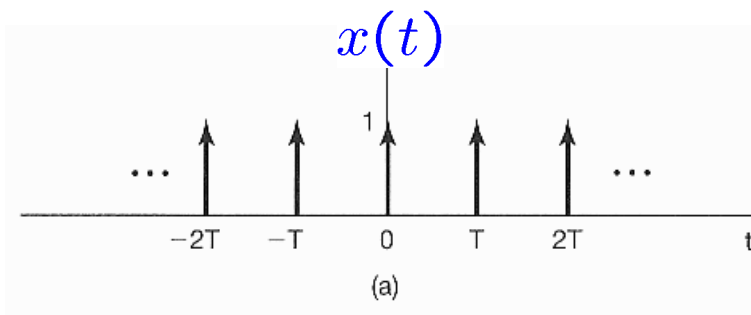
$$\Rightarrow a_k = \frac{1}{T} \int_{-T/2}^{+T/2} \delta(t) e^{-jk\omega_0 t} dt = \frac{1}{T}$$

$$\Rightarrow X(j\omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta(\omega - \frac{2\pi}{T}k)$$

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$$

$$a_k = \frac{1}{T} \int_{-T/2}^{+T/2} \tilde{x}(t) e^{-jk\omega_0 t} dt$$

$$X(j\omega) = \sum_{k=-\infty}^{+\infty} 2\pi a_k \delta(\omega - k\omega_0)$$



- Representation of **Aperiodic** Signals:
the Continuous-Time Fourier Transform
- The Fourier Transform for **Periodic** Signals
- **Properties**
of the Continuous-Time Fourier Transform
- The Convolution Property
- The Multiplication Property
- Systems Characterized by Linear Constant-Coefficient Differential Equations

Section	Property
4.3.1	Linearity
4.3.2	Time Shifting
4.3.6	Frequency Shifting
4.3.3	Conjugation
4.3.5	Time Reversal
4.3.5	Time and Frequency Scaling
4.4	Convolution
4.5	Multiplication
4.3.4	Differentiation in Time
4.3.4	Integration
4.3.6	Differentiation in Frequency
4.3.3	Conjugate Symmetry for Real Signals
4.3.3	Symmetry for Real and Even Signals
4.3.3	Symmetry for Real and Odd Signals
4.3.3	Even-Odd Decomposition for Real Signals
4.3.7	Parseval's Relation for Aperiodic Signals

Property	CTFS	DTFS	CTFT	DTFT	LT	zT
Linearity	3.5.1		4.3.1	5.3.2	9.5.1	10.5.1
Time Shifting	3.5.2		4.3.2	5.3.3	9.5.2	10.5.2
Frequency Shifting (in s, z)			4.3.6	5.3.3	9.5.3	10.5.3
Conjugation	3.5.6		4.3.3	5.3.4	9.5.5	10.5.6
Time Reversal	3.5.3		4.3.5	5.3.6		10.5.4
Time & Frequency Scaling	3.5.4		4.3.5	5.3.7	9.5.4	10.5.5
(Periodic) Convolution			4.4	5.4	9.5.6	10.5.7
Multiplication	3.5.5	3.7.2	4.5	5.5		
Differentiation/First Difference		3.7.2	4.3.4, 4.3.6	5.3.5, 5.3.8	9.5.7, 9.5.8	10.5.7, 10.5.8
Integration/Running Sum (Accumulation)			4.3.4	5.3.5	9.5.9	10.5.7
Conjugate Symmetry for Real Signals	3.5.6		4.3.3	5.3.4		
Symmetry for Real and Even Signals	3.5.6		4.3.3	5.3.4		
Symmetry for Real and Odd Signals	3.5.6		4.3.3	5.3.4		
Even-Odd Decomposition for Real Signals			4.3.3	5.3.4		
Parseval's Relation for (A)Periodic Signals	3.5.7	3.7.3	4.3.7	5.3.9		
Initial- and Final-Value Theorems					9.5.10	10.5.9

■ Fourier Transform Pair:

- Synthesis equation: $x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$

- Analysis equation: $X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$

- Notations:

$$X(j\omega) = \mathcal{F}\{x(t)\}$$

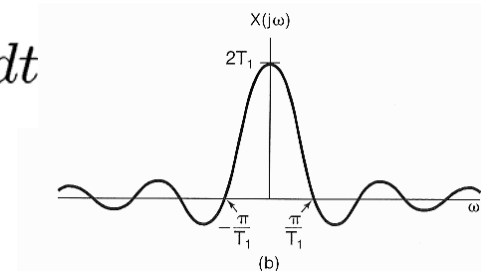
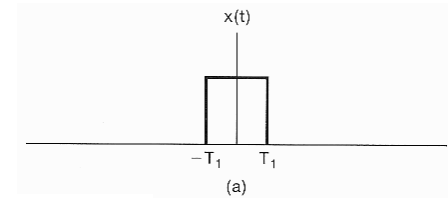
$$\frac{1}{a + j\omega} = \mathcal{F}\{e^{-at}u(t)\}$$

$$x(t) = \mathcal{F}^{-1}\{X(j\omega)\}$$

$$e^{-at}u(t) = \mathcal{F}^{-1}\left\{\frac{1}{a + j\omega}\right\}$$

$$x(t) \xleftrightarrow{CT\mathcal{F}T} X(j\omega)$$

$$e^{-at}u(t) \xleftrightarrow{CT\mathcal{F}T} \frac{1}{a + j\omega}$$



■ Linearity:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$$

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$$

$$x(t) \xleftrightarrow{\mathcal{F}} X(j\omega)$$

$$y(t) \xleftrightarrow{\mathcal{F}} Y(j\omega)$$

$$\Rightarrow a x(t) + b y(t) \xleftrightarrow{\mathcal{F}} a X(j\omega) + b Y(j\omega)$$

■ Time Shifting:

$$x(t) \xleftrightarrow{\mathcal{F}} X(j\omega)$$

$$\Rightarrow x(t - t_0) \xleftrightarrow{\mathcal{F}} e^{-j\omega t_0} X(j\omega)$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$$

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$$

$$Y(j\omega) = \int_{-\infty}^{+\infty} x(t - t_0) e^{-j\omega t} dt$$

$$x(t - t_0) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega(t - t_0)} d\omega$$

$$= \int_{-\infty}^{+\infty} x(\tau) e^{-j\omega(\tau + t_0)} d\tau$$

$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \left(e^{-j\omega t_0} X(j\omega) \right) e^{j\omega t} d\omega$$

$$= e^{-j\omega t_0} \int_{-\infty}^{+\infty} x(\tau) e^{-j\omega \tau} d\tau$$

■ Time Shift → Phase Shift:

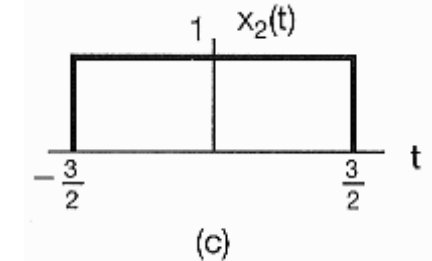
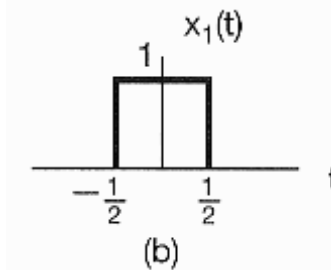
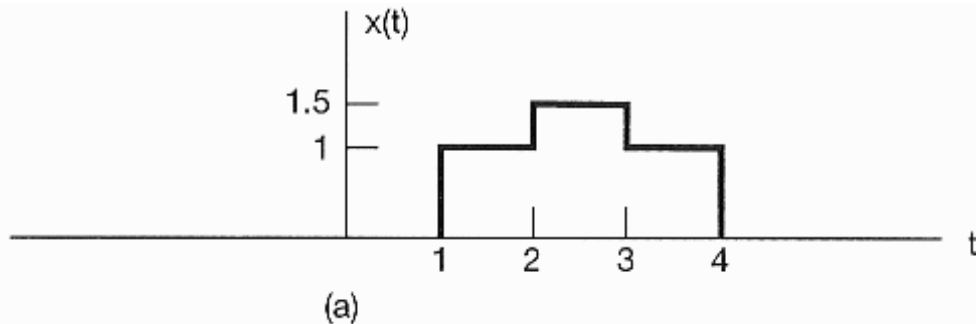
$$\mathcal{F}\{x(t)\} = X(j\omega) = |X(j\omega)|e^{j\angle X(j\omega)}$$

$$\mathcal{F}\{x(t-t_0)\} = e^{-j\omega t_0} X(j\omega) = |X(j\omega)|e^{j[\angle X(j\omega) - \omega t_0]}$$

■ Example 4.9:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$$

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$$



$$x(t) = \frac{1}{2} x_1(t - 2.5) + x_2(t - 2.5)$$

$$X_1(j\omega) = \frac{2 \sin(\omega/2)}{\omega}$$

$$X_2(j\omega) = \frac{2 \sin(3\omega/2)}{\omega}$$

$$\Rightarrow X(j\omega) = e^{-j5\omega/2} \left\{ \frac{\sin(\omega/2) + 2 \sin(3\omega/2)}{\omega} \right\}$$

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$$

■ Conjugation & Conjugate Symmetry:

$$x(t) \xleftrightarrow{\mathcal{F}} X(j\omega)$$

$$x(t)^* \xleftrightarrow{\mathcal{F}} X^*(-j\omega)$$

$$\begin{aligned} x(t) &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\bar{\omega}) e^{j\bar{\omega} t} d\bar{\omega} \\ &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega \end{aligned}$$

■ Conjugation & Conjugate Symmetry:

$$x(t) \xleftrightarrow{\mathcal{F}} X(j\omega)$$

$$x(t)^* \xleftrightarrow{\mathcal{F}} X^*(-j\omega)$$

$$\bullet x(t) = x^*(t) \Rightarrow X(-j\omega) = X^*(j\omega)$$

$x(t)$ is real $\Rightarrow X(j\omega)$ is conjugate symmetric

$$\bullet x(t) = x^*(t) \ \& \ x(-t) = x(t)$$

$$\Rightarrow X(-j\omega) = X^*(j\omega) \ \& \ X(-j\omega) = X(j\omega)$$

$$\Rightarrow X(j\omega) = X^*(j\omega)$$

$x(t)$ is real & even $\Rightarrow X(j\omega)$ are real & even

$$\bullet x(t) \text{ is real \& odd} \Rightarrow X(j\omega) \text{ are purely imaginary \& odd}$$

■ Conjugation & Conjugate Symmetry:

If $x(t)$ is a **real** function

$$x(t) = \mathcal{E}v\{x(t)\} + \mathcal{O}d\{x(t)\} = x_e(t) + x_o(t)$$

$$\Rightarrow \mathcal{F}\{x(t)\} = \mathcal{F}\{x_e(t)\} + \mathcal{F}\{x_o(t)\}$$

$$\Rightarrow \mathcal{F}\{x_e(t)\} : \text{a **real** function}$$

$$\Rightarrow \mathcal{F}\{x_o(t)\} : \text{a **purely imaginary** function}$$

$$x(t) \xleftrightarrow{\mathcal{F}} X(jw)$$

$$\mathcal{E}v\{x(t)\} \xleftrightarrow{\mathcal{F}} \mathcal{R}e\{X(jw)\}$$

$$\mathcal{O}d\{x(t)\} \xleftrightarrow{\mathcal{F}} j \mathcal{I}m\{X(jw)\}$$

■ Example 4.10:

Ex 4.1

$$e^{-at}u(t) \xleftrightarrow{\mathcal{F}} \frac{1}{a + jw}$$

Ex 4.2

$$e^{-a|t|} \xleftrightarrow{\mathcal{F}} ?$$

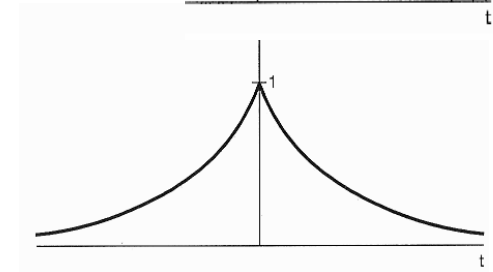
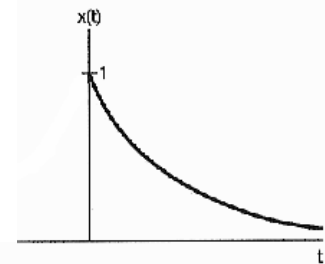
$$x(t) = e^{-a|t|} = e^{-at}u(t) + e^{at}u(-t)$$

$$= 2 \left[\frac{e^{-at}u(t) + e^{at}u(-t)}{2} \right] = 2\mathcal{E}v \{ e^{-at}u(t) \}$$

$$\mathcal{E}v \{ e^{-at}u(t) \} \xleftrightarrow{\mathcal{F}} \mathcal{R}e \left\{ \frac{1}{a + jw} \right\}$$

$$\mathcal{O}d \{ e^{-at}u(t) \} \xleftrightarrow{\mathcal{F}} j \mathcal{I}m \left\{ \frac{1}{a + jw} \right\}$$

$$X(jw) = 2\mathcal{R}e \left\{ \frac{1}{a + jw} \right\} = \frac{2a}{a^2 + w^2}$$



■ Differentiation & Integration:

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$$

$$x(t) \xleftrightarrow{\mathcal{F}} X(j\omega)$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$$

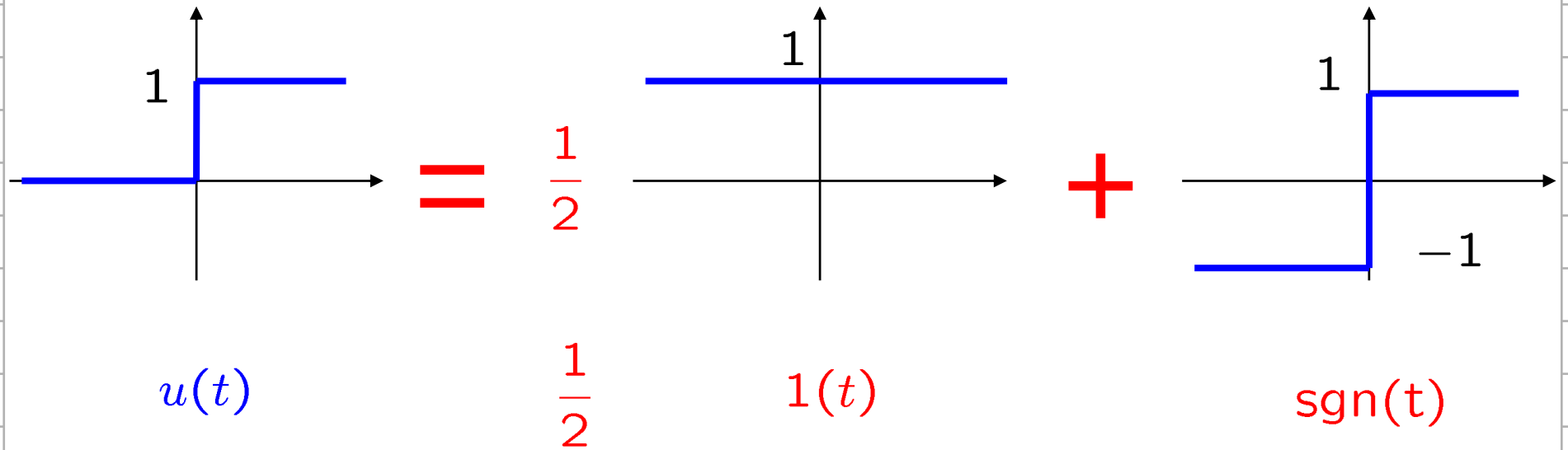
$$\frac{d}{dt} x(t) \xleftrightarrow{\mathcal{F}} j\omega X(j\omega)$$

$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$$

$$\int_{-\infty}^t x(\tau) d\tau \xleftrightarrow{\mathcal{F}} \frac{1}{j\omega} X(j\omega) + \pi X(0) \delta(\omega)$$

dc or average value



$$1 \xleftrightarrow{\mathcal{FT}} 2\pi\delta(j\omega) \quad \text{sgn}(t) \xleftrightarrow{\mathcal{FT}} S(j\omega)$$

$$\frac{d}{dt} \text{sgn}(t) \xleftrightarrow{\mathcal{FT}} j\omega S(j\omega)$$

$$2\delta(t) \xleftrightarrow{\mathcal{FT}} j\omega S(j\omega)$$

$$\delta(t) \xleftrightarrow{\mathcal{FT}} 1$$

$$\Rightarrow U(j\omega) =$$

$$\Rightarrow S(j\omega) =$$

■ Example 4.11:

$$x(t) = u(t) \xleftrightarrow{\mathcal{F}} X(j\omega) = ?$$

$$g(t) = \delta(t) \xleftrightarrow{\mathcal{F}} G(j\omega) = 1$$

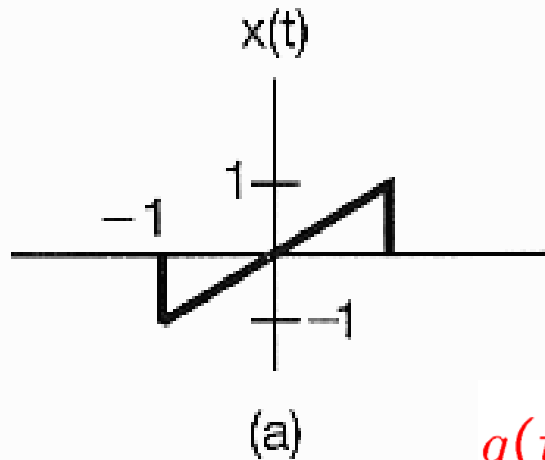
$$x(t) = \int_{-\infty}^t g(\tau) d\tau$$

$$X(j\omega) = \frac{1}{j\omega} G(j\omega) + \pi G(0) \delta(\omega)$$

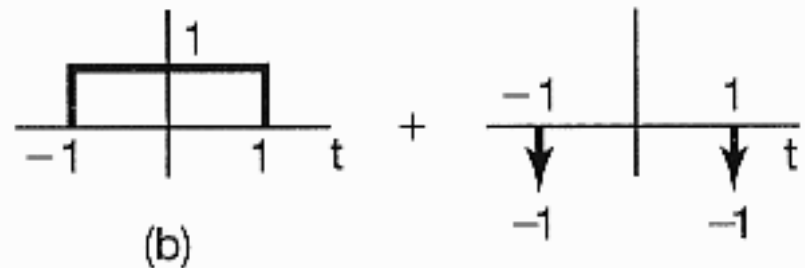
$$= \frac{1}{j\omega} + \pi \delta(\omega)$$

$$\delta(t) = \frac{d}{dt} u(t) \xleftrightarrow{\mathcal{F}} j\omega \left[\frac{1}{j\omega} + \pi \delta(\omega) \right] = 1$$

■ Example 4.12:



$$g(t) = \frac{dx(t)}{dt}$$



$$g(t) = \frac{d}{dt}x(t)$$

$$G(j\omega) = \frac{2 \sin(\omega)}{\omega} - e^{j\omega} - e^{-j\omega}$$

$$\Rightarrow X(j\omega) = \frac{G(j\omega)}{j\omega} + \pi G(0)\delta(\omega)$$

$$= \frac{2 \sin(\omega)}{j\omega^2} - \frac{2 \cos(\omega)}{j\omega}$$

Time & Frequency Scaling:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$$

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$$

$$x(t) \xleftrightarrow{\mathcal{F}} X(j\omega)$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$$

$$x(at) \xleftrightarrow{\mathcal{F}} \frac{1}{|a|} X\left(\frac{j\omega}{a}\right)$$

$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\bar{\omega}) e^{j\bar{\omega} t} d\bar{\omega}$$

$$\frac{1}{|b|} x\left(\frac{t}{b}\right) \xleftrightarrow{\mathcal{F}} X(jb\omega)$$

$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\bar{\omega}) e^{j\bar{\omega} t} d\bar{\omega}$$

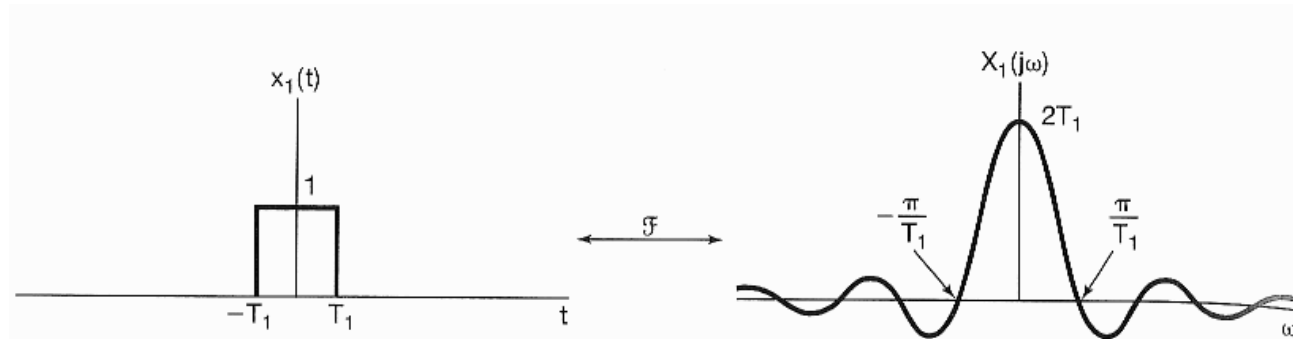
$$x(-t) \xleftrightarrow{\mathcal{F}} X(-j\omega)$$

$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\bar{\omega}) e^{j\bar{\omega} t} d\bar{\omega}$$

$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\bar{\omega}) e^{j\bar{\omega} t} d\bar{\omega}$$

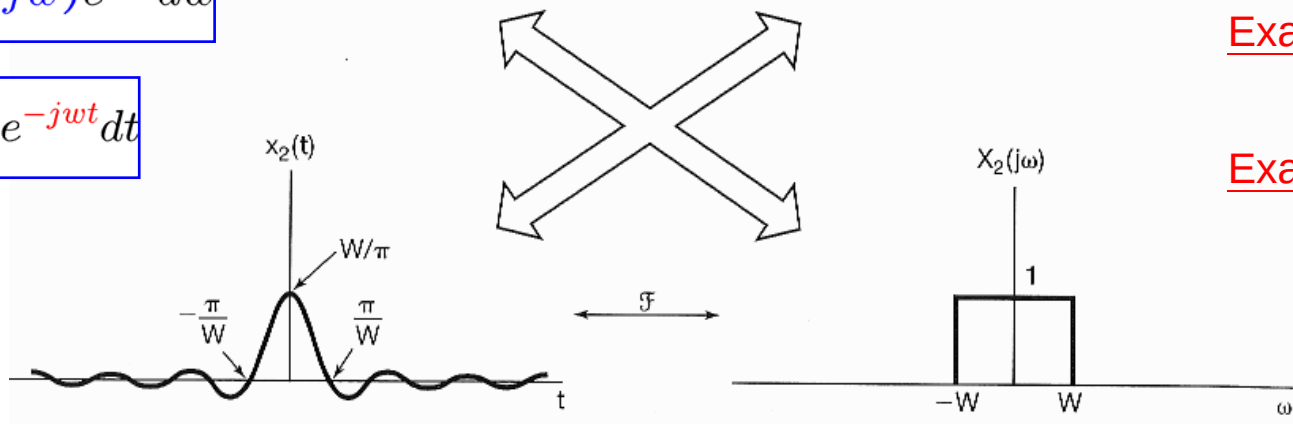
■ Duality:

$$x_1(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & |t| > T_1 \end{cases} \xleftrightarrow{\mathcal{F}} X_1(j\omega) = \frac{2 \sin(\omega T_1)}{\omega}$$



$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$$

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$$



Example 4.4

Example 4.5

$$x_2(t) = \frac{\sin(Wt)}{\pi t} \xleftrightarrow{\mathcal{F}} X_2(j\omega) = \begin{cases} 1, & |\omega| < W \\ 0, & |\omega| > W \end{cases}$$

■ Duality:

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$$

$$B(s) = \int_{-\infty}^{+\infty} A(\tau) e^{-js\tau} d\tau$$

$$B(-s) = \int_{-\infty}^{+\infty} A(\tau) e^{js\tau} d\tau$$

$$A(\tau) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} B(s) e^{js\tau} ds$$

$$A(s) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} B(\tau) e^{js\tau} d\tau$$

$$A(-s) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} B(\tau) e^{-js\tau} d\tau$$

■ Duality:

$$x(t) \xleftrightarrow{\mathcal{F}} X(j\omega)$$

$$x(t - t_0) \xleftrightarrow{\mathcal{F}} e^{-j\omega t_0} X(j\omega)$$

$$\frac{d}{dt}x(t) \xleftrightarrow{\mathcal{F}} j\omega X(j\omega)$$

$$\int_{-\infty}^t x(\tau) d\tau \xleftrightarrow{\mathcal{F}} \frac{1}{j\omega} X(j\omega) + \pi X(0) \delta(\omega)$$

$$-jtx(t) \xleftrightarrow{\mathcal{F}} \frac{d}{d\omega} X(j\omega)$$

$$e^{j\omega_0 t} x(t) \xleftrightarrow{\mathcal{F}} X(j(\omega - \omega_0))$$

$$-\frac{1}{jt} x(t) + \pi x(0) \delta(t) \xleftrightarrow{\mathcal{F}} \int_{-\infty}^{\omega} X(\eta) d\eta$$

■ Parseval's relation:

$$x(t) \xleftrightarrow{\mathcal{F}} X(j\omega)$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$$

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$$

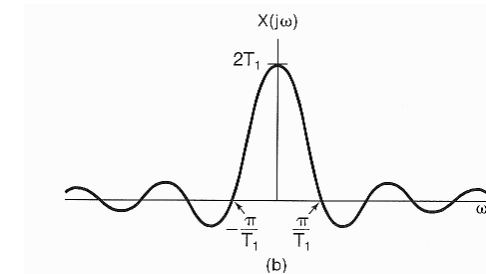
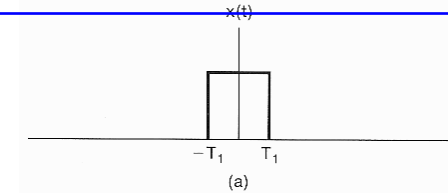
$$\Rightarrow \int_{-\infty}^{+\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} |X(j\omega)|^2 d\omega$$

$$\int_{-\infty}^{+\infty} |x(t)|^2 dt = \int_{-\infty}^{+\infty} x(t) x^*(t) dt$$

$$= \int_{-\infty}^{+\infty} x(t) \left[\frac{1}{2\pi} \int_{-\infty}^{+\infty} X^*(j\omega) e^{-j\omega t} d\omega \right] dt$$

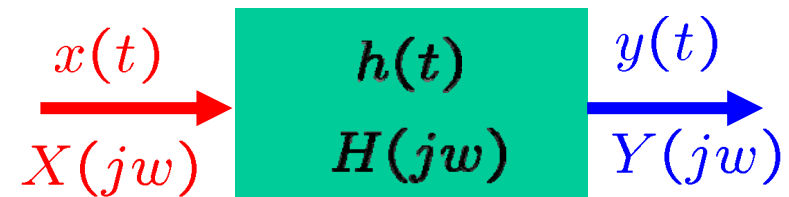
$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} X^*(j\omega) \left[\int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt \right] d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} |X(j\omega)|^2 d\omega$$



- Representation of **Aperiodic** Signals:
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- **The Multiplication Property**
- Systems Characterized by
Linear Constant-Coefficient Differential Equations

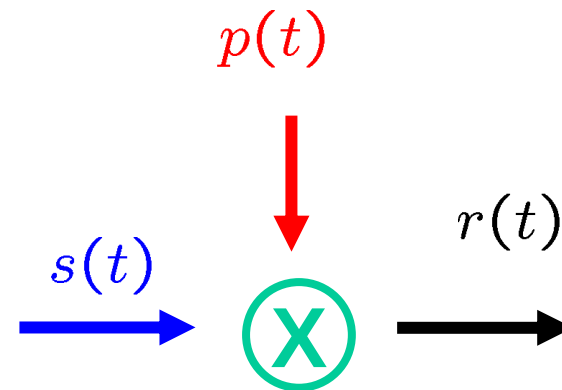
■ Convolution Property:



$$y(t) = x(t) * h(t) \xleftrightarrow{\mathcal{F}} Y(jw) = X(jw)H(jw)$$

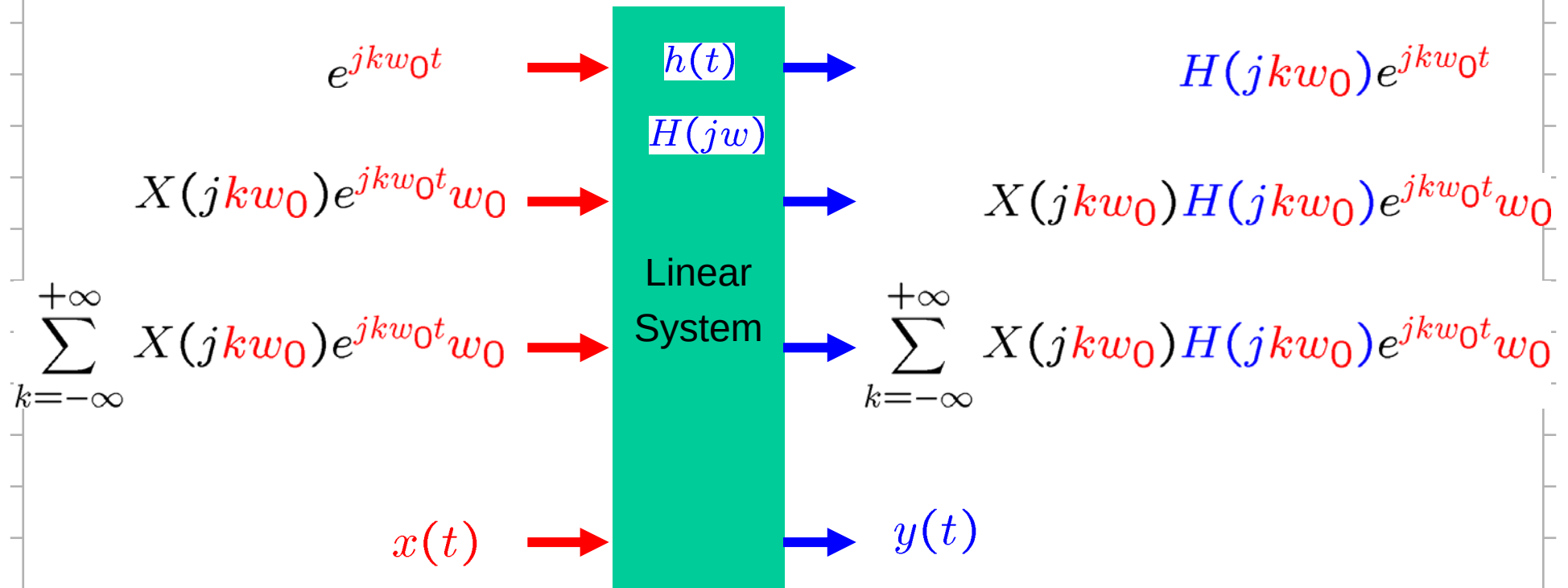
$$= \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau$$

■ Multiplication Property:



$$r(t) = s(t)p(t) \xleftrightarrow{\mathcal{F}} R(jw) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S(j\theta)P(j(w - \theta))d\theta$$

- From Superposition (or Linearity): $H(jk\omega_0) = \int_{-\infty}^{\infty} h(t)e^{-jk\omega_0 t} dt$



$$= \lim_{\omega_0 \rightarrow 0} \frac{1}{2\pi} \sum_{k=-\infty}^{+\infty} X(jk\omega_0)e^{jk\omega_0 t}\omega_0$$

$$= \lim_{\omega_0 \rightarrow 0} \frac{1}{2\pi} \sum_{k=-\infty}^{+\infty} X(jk\omega_0)H(jk\omega_0)e^{jk\omega_0 t}\omega_0$$

$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega)e^{j\omega t}d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega)H(j\omega)e^{j\omega t}d\omega$$

■ From Superposition (or Linearity):

$$\frac{1}{2\pi} \sum_{k=-\infty}^{+\infty} X(jkw_0) e^{jkw_0 t} w_0 \longrightarrow \frac{1}{2\pi} \sum_{k=-\infty}^{+\infty} X(jkw_0) H(jkw_0) e^{jkw_0 t} w_0$$

$$y(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(jw) H(jw) e^{jw t} dw$$

Since $y(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} Y(jw) e^{jw t} dw$

$$\Rightarrow Y(jw) = X(jw) H(jw)$$

$$y(t) = x(t) * h(t) \xleftrightarrow{\mathcal{F}} Y(jw) = X(jw) H(jw)$$

■ From Convolution Integral:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(jw) e^{jw t} dw$$

$$X(jw) = \int_{-\infty}^{+\infty} x(t) e^{-jw t} dt$$

$$x(t - t_0) \xleftrightarrow{\mathcal{F}} e^{-jw t_0} X(jw)$$

$$y(t) = \int_{-\infty}^{+\infty} x(\tau) h(t - \tau) d\tau$$

$$\Rightarrow Y(jw) = \mathcal{F}\{y(t)\} = \int_{-\infty}^{+\infty} \left[\int_{-\infty}^{+\infty} x(\tau) h(t - \tau) d\tau \right] e^{-jw t} dt$$

$$= \int_{-\infty}^{+\infty} x(\tau) \left[\int_{-\infty}^{+\infty} h(t - \tau) e^{-jw t} dt \right] d\tau$$

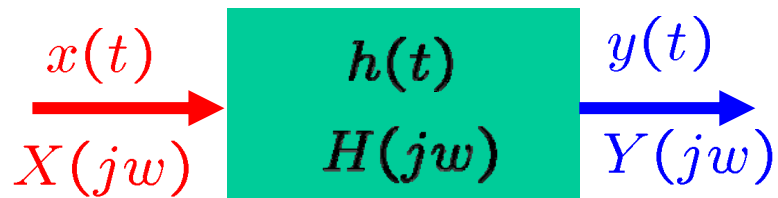
$$= \int_{-\infty}^{+\infty} x(\tau) \left[e^{-jw \tau} \int_{-\infty}^{+\infty} h(\sigma) e^{-jw \sigma} d\sigma \right] d\tau$$

$$= \int_{-\infty}^{+\infty} x(\tau) \left[e^{-jw \tau} H(jw) \right] d\tau$$

$$= H(jw) \int_{-\infty}^{+\infty} x(\tau) e^{-jw \tau} d\tau$$

$$\Rightarrow Y(jw) = H(jw) X(jw)$$

■ Equivalent LTI Systems:



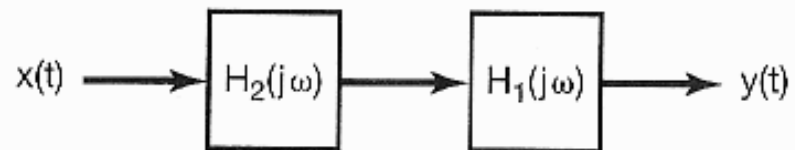
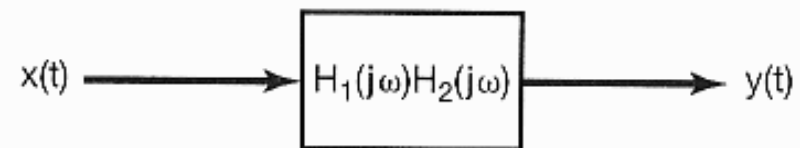
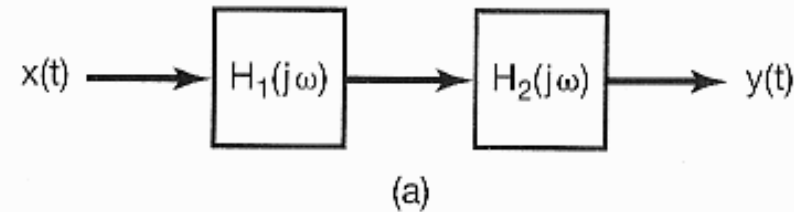
$$h(t) \xleftrightarrow{\mathcal{F}} H(j\omega)$$

impulse
response

frequency
response

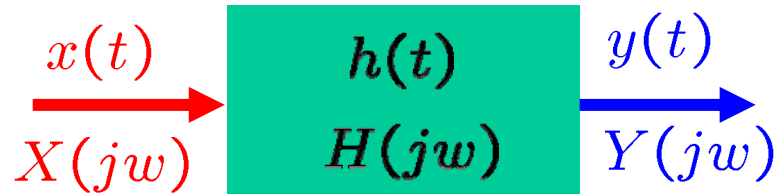
$$y(t) = x(t) * h(t)$$

$$Y(j\omega) = X(j\omega)H(j\omega)$$



$$\Rightarrow Y(j\omega) = H_1(j\omega)H_2(j\omega)X(j\omega)$$

■ Example 4.15: Time Shift



$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(jw) e^{jw t} dw$$

$$X(jw) = \int_{-\infty}^{+\infty} x(t) e^{-jw t} dt$$

$$x(t - t_0) \xleftrightarrow{\mathcal{F}} e^{-jw t_0} X(jw)$$

$$h(t) = \delta(t - t_0)$$

$$\Rightarrow H(jw) = e^{-jw t_0}$$

$$Y(jw) = H(jw) X(jw)$$

$$= e^{-jw t_0} X(jw)$$

$$\Rightarrow y(t) = x(t - t_0)$$

■ Examples 4.16 & 17: Differentiator & Integrator

$$y(t) = \frac{d}{dt}x(t) \quad \Rightarrow \quad Y(j\omega) = j\omega X(j\omega)$$

$$\Rightarrow H(j\omega) = j\omega$$

$$y(t) = \int_{-\infty}^t x(\tau) d\tau \quad \Rightarrow \quad h(t) = u(t) \quad \text{impulse response}$$

$$\Rightarrow H(j\omega) = \frac{1}{j\omega} + \pi\delta(\omega)$$

$$\Rightarrow Y(j\omega) = H(j\omega)X(j\omega)$$

$$= \frac{1}{j\omega}X(j\omega) + \pi\delta(\omega)X(j\omega)$$

$$= \frac{1}{j\omega}X(j\omega) + \pi\delta(\omega)X(0)$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$$

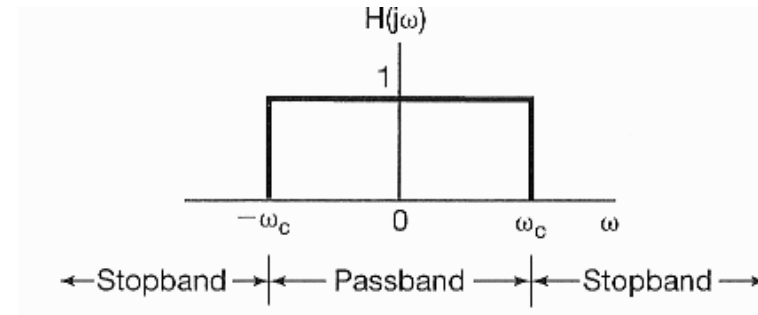
$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$$

■ Example 4.18: Ideal Lowpass Filter

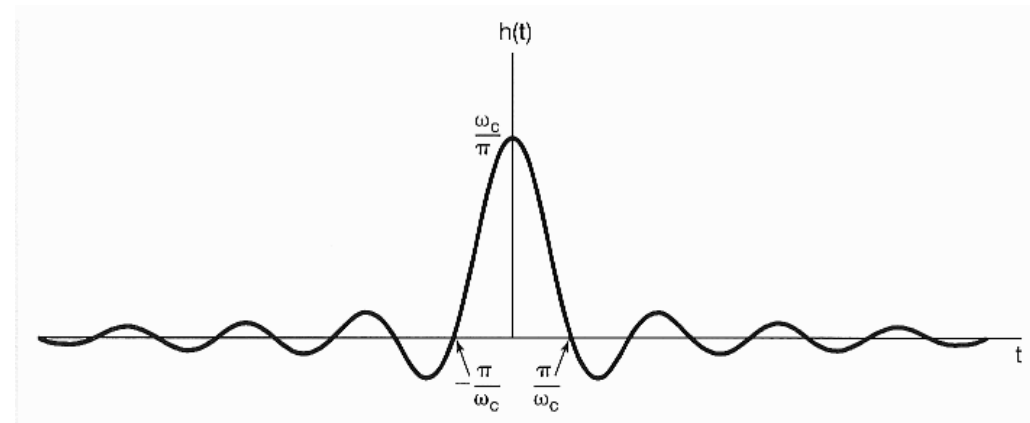
$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$$

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$$

$$H(j\omega) = \begin{cases} 1, & |\omega| < \omega_c \\ 0, & |\omega| > \omega_c \end{cases}$$

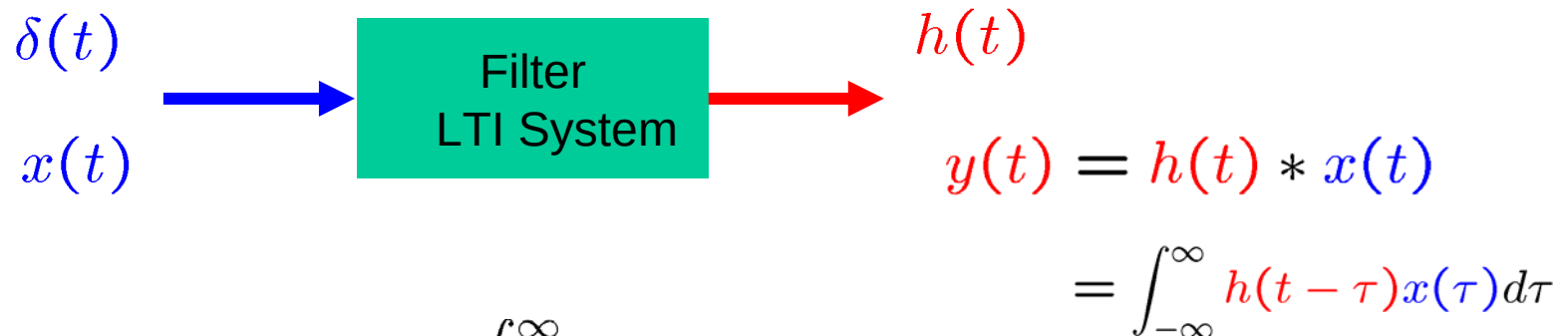


$$\begin{aligned} \Rightarrow h(t) &= \frac{1}{2\pi} \int_{-\omega_c}^{+\omega_c} e^{j\omega t} d\omega \\ &= \frac{\sin(\omega_c t)}{\pi t} \end{aligned}$$



■ Filter Design:

$$H(j\omega) = \int_{-\infty}^{\infty} h(t) e^{-j\omega t} dt$$



$$H(j\omega) = \int_{-\infty}^{\infty} h(t) e^{-j\omega t} dt$$

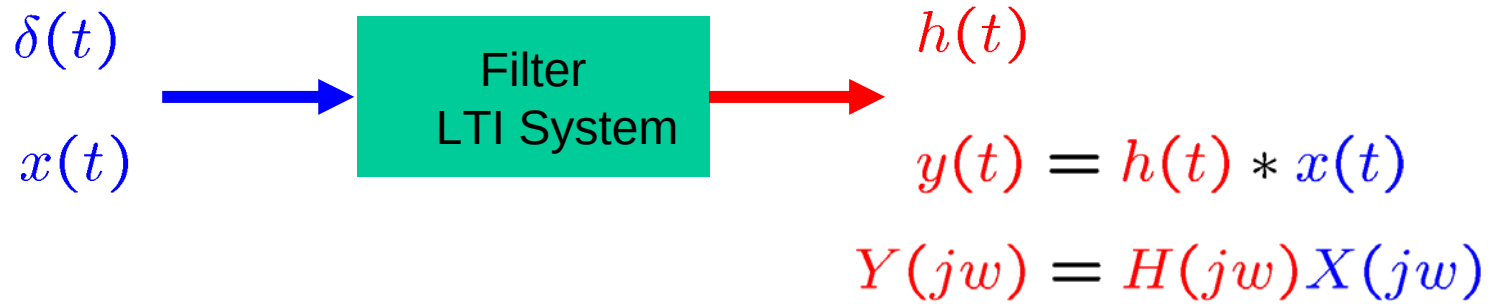
$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$\Rightarrow Y(j\omega) = H(j\omega) X(j\omega)$$

$$\Rightarrow y(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} Y(j\omega) e^{j\omega t} d\omega$$

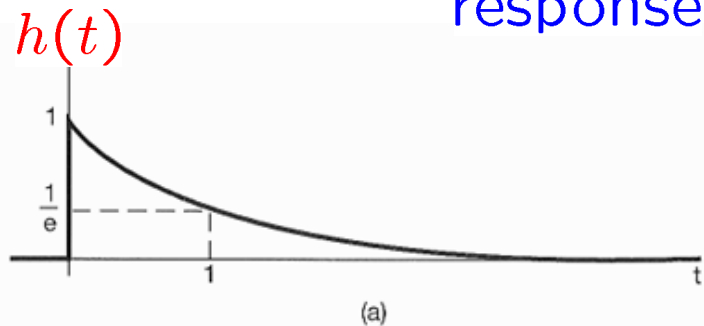
Filter Design:

$$H(j\omega) = \int_{-\infty}^{\infty} h(t)e^{-j\omega t} dt$$

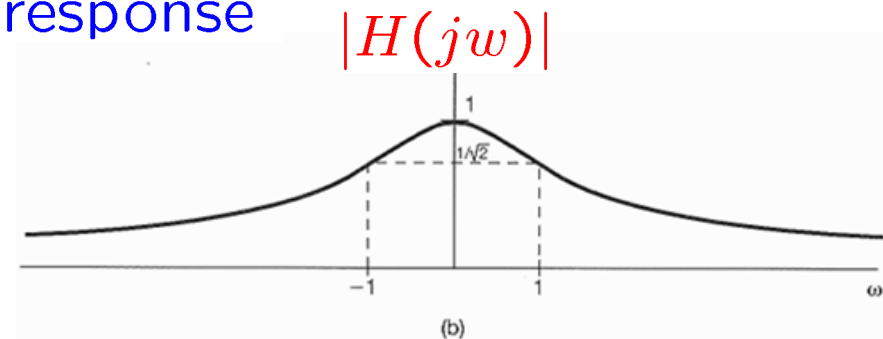


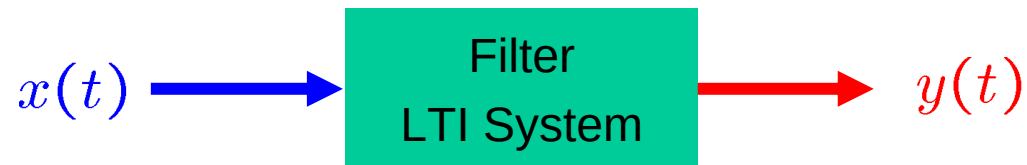
RC circuit

$$h(t) = e^{-t}u(t) \xleftrightarrow{\mathcal{F}} H(j\omega) = \frac{1}{j\omega + 1}$$



frequency response



■ Example 4.19:

$$h(t) = e^{-at}u(t), \quad a > 0 \quad \Rightarrow \quad H(j\omega) = \frac{1}{a + j\omega}$$

$$x(t) = e^{-bt}u(t), \quad b > 0 \quad \Rightarrow \quad X(j\omega) = \frac{1}{b + j\omega}$$

$$\Rightarrow Y(j\omega) = H(j\omega)X(j\omega) = \frac{1}{a + j\omega} \frac{1}{b + j\omega}$$

$$\text{if } a \neq b \quad = \frac{1}{b - a} \left[\frac{1}{a + j\omega} - \frac{1}{b + j\omega} \right]$$

■ Example 4.19:

$$\text{if } a \neq b \quad Y(j\omega) = \frac{1}{b-a} \left[\frac{1}{a+j\omega} - \frac{1}{b+j\omega} \right]$$
$$\Rightarrow y(t) = \frac{1}{b-a} \left[e^{-at}u(t) - e^{-bt}u(t) \right]$$

$$\text{if } a = b \quad Y(j\omega) = \frac{1}{(a+j\omega)^2}$$

$$\text{since } e^{-at}u(t) \xleftrightarrow{\mathcal{F}} \frac{1}{a+j\omega}$$

$$\text{and } t e^{-at}u(t) \xleftrightarrow{\mathcal{F}} j \frac{d}{d\omega} \left[\frac{1}{a+j\omega} \right] = \frac{1}{(a+j\omega)^2}$$

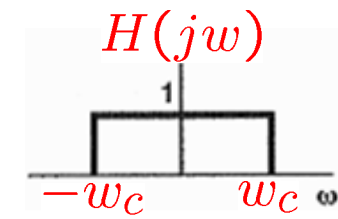
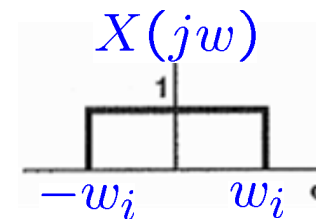
$$\Rightarrow y(t) = t e^{-at}u(t)$$

■ Example 4.20: $h(t) = \frac{\sin(\omega_c t)}{\pi t}$ $H(j\omega) = \int_{-\infty}^{\infty} h(t)e^{-j\omega t} dt$

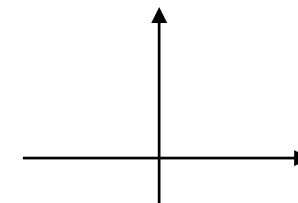


$\Rightarrow Y(j\omega) = H(j\omega)X(j\omega)$

$\Rightarrow X(j\omega) = \begin{cases} 1, & |\omega| \leq \omega_i \\ 0, & \text{otherwise} \end{cases}$



$\Rightarrow H(j\omega) = \begin{cases} 1, & |\omega| \leq \omega_c \\ 0, & \text{otherwise} \end{cases}$



$\Rightarrow Y(j\omega) = \begin{cases} 1, & |\omega| \leq \omega_0 \\ 0, & \text{otherwise} \end{cases}$
 $\omega_0 = \min(\omega_c, \omega_i)$

$\Rightarrow y(t) = \frac{\sin(\omega_0 t)}{\pi t}$

$\Rightarrow y(t) = \begin{cases} \frac{\sin(\omega_c t)}{\pi t}, & \omega_c \leq \omega_i \\ \frac{\sin(\omega_i t)}{\pi t}, & \omega_c \geq \omega_i \end{cases}$

- Representation of **Aperiodic** Signals:
the Continuous-Time Fourier Transform
- The Fourier Transform for **Periodic** Signals
- **Properties** of the Continuous-Time Fourier Transform
- The **Convolution** Property
- **The Multiplication Property**
- Systems Characterized by
Linear Constant-Coefficient Differential Equations

■ Convolution & Multiplication:

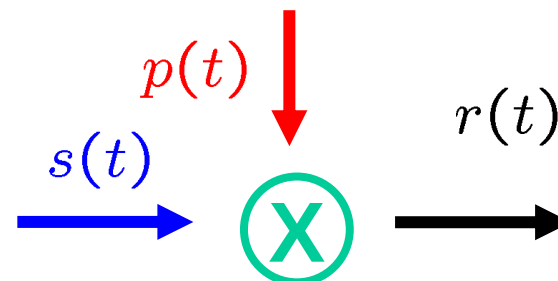
$$y(t) = x(t) * h(t) \xleftrightarrow{\mathcal{F}} Y(j\omega) = X(j\omega)H(j\omega)$$

$$= \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau$$

$$r(t) = s(t)p(t) \xleftrightarrow{\mathcal{F}} R(j\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S(j\theta)P(j(\omega - \theta))d\theta$$

■ Multiplication of One Signal by Another:

- Scale or modulate the amplitude of the other signal
- Modulation



Multiplication Property

$$r(t) = s(t)p(t)$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(jw) e^{jw t} dw$$

$$X(jw) = \int_{-\infty}^{+\infty} x(t) e^{-jw t} dt$$

$$\Rightarrow R(jw) = \int_{-\infty}^{\infty} r(t) e^{-jw t} dt$$

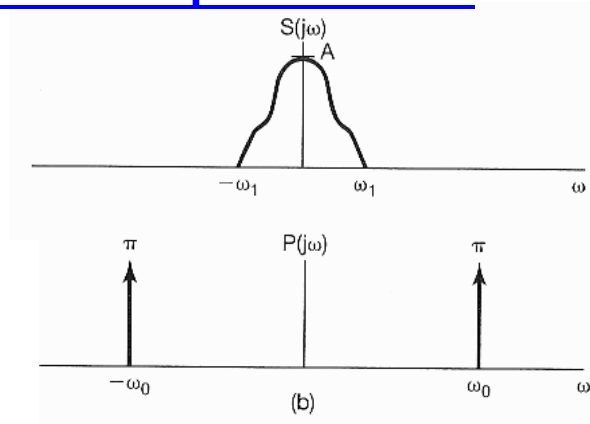
$$= \int_{-\infty}^{\infty} s(t) p(t) e^{-jw t} dt$$

$$= \int_{-\infty}^{\infty} s(t) \left\{ \frac{1}{2\pi} \int_{-\infty}^{\infty} P(j\theta) e^{j\theta t} d\theta \right\} e^{-jw t} dt$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} P(j\theta) \left[\int_{-\infty}^{\infty} s(t) e^{-j(w-\theta)t} dt \right] d\theta$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} P(j\theta) S(j(w-\theta)) d\theta = \frac{1}{2\pi} \int_{-\infty}^{\infty} P(j(w-\theta)) S(j\theta) d\theta$$

■ Example 4.21:



$$r(t) = s(t)p(t)$$

$$s(t) \xleftrightarrow{\mathcal{F}} S(j\omega)$$

$$p(t) \xleftrightarrow{\mathcal{F}} P(j\omega)$$

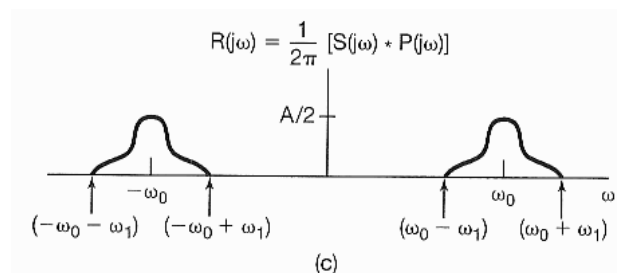
$$p(t) = \cos(\omega_0 t)$$

$$P(j\omega) = \pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0)$$

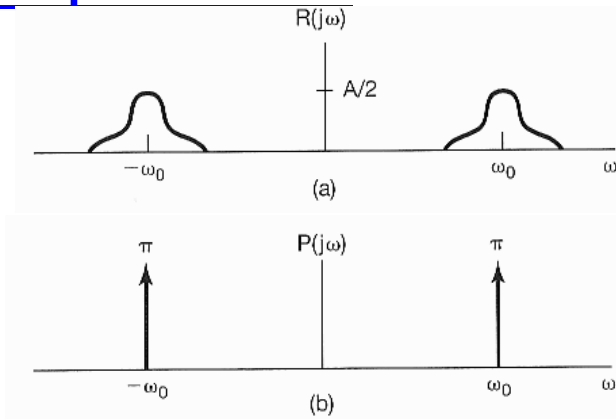
$$R(j\omega) = \frac{1}{2\pi} [S(j\omega) * P(j\omega)]$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} S(j\theta) P(j(\omega - \theta)) d\theta$$

$$= \frac{1}{2} S(j(\omega - \omega_0)) + \frac{1}{2} S(j(\omega + \omega_0))$$



■ Example 4.22:



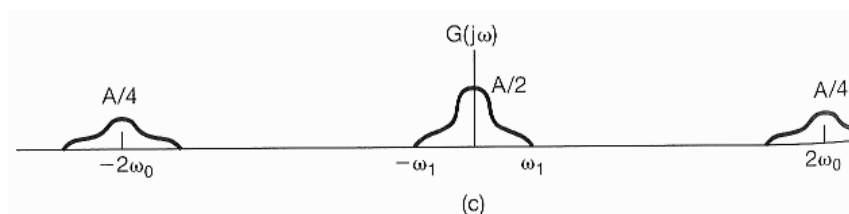
$$g(t) = r(t)p(t)$$

$$r(t) \xleftrightarrow{\mathcal{F}} R(j\omega)$$

$$p(t) \xleftrightarrow{\mathcal{F}} P(j\omega)$$

$$p(t) = \cos(\omega_0 t)$$

$$G(j\omega) = \frac{1}{2\pi} [R(j\omega) * P(j\omega)]$$



■ Example 4.23:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$$

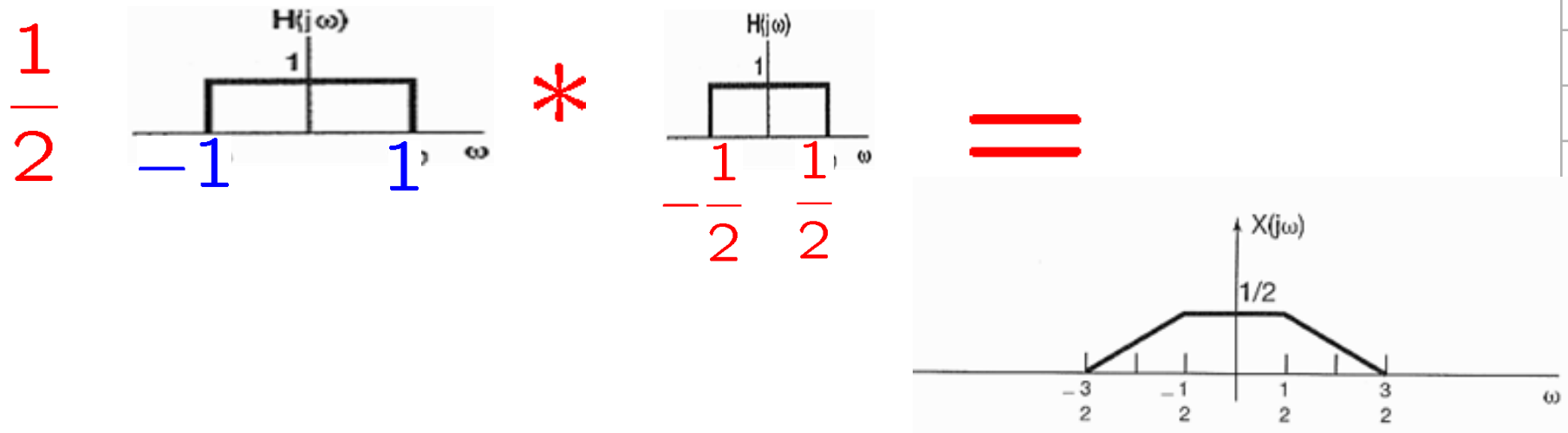
$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$$

$$x(t) = \frac{\sin(t) \sin(t/2)}{\pi t^2}$$

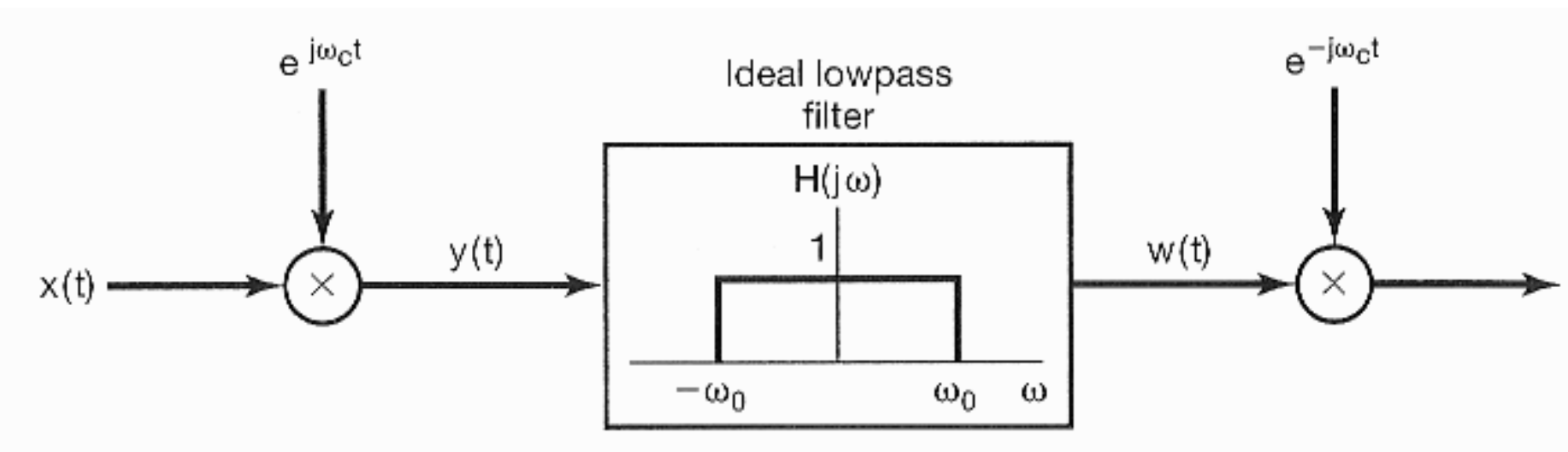
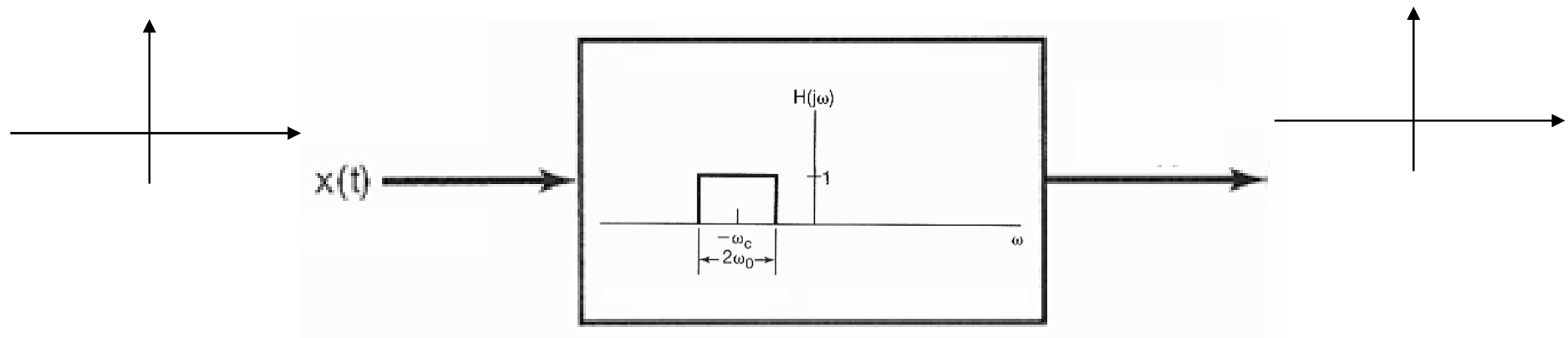
$$X(j\omega) = \int_{-\infty}^{\infty} \frac{\sin(t) \sin(t/2)}{\pi t^2} e^{-j\omega t} dt$$

$$= \pi \left(\frac{\sin(t)}{\pi t} \right) \left(\frac{\sin(t/2)}{\pi t} \right)$$

$$X(j\omega) = \frac{1}{2} \mathcal{F} \left\{ \frac{\sin(t)}{\pi t} \right\} * \mathcal{F} \left\{ \frac{\sin(t/2)}{\pi t} \right\}$$

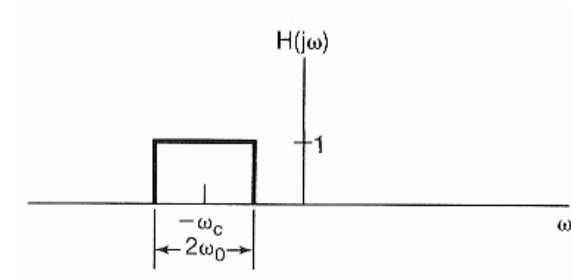
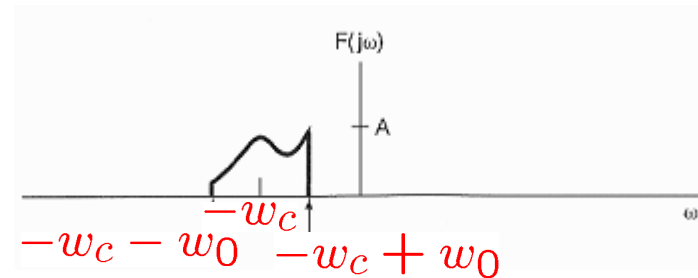
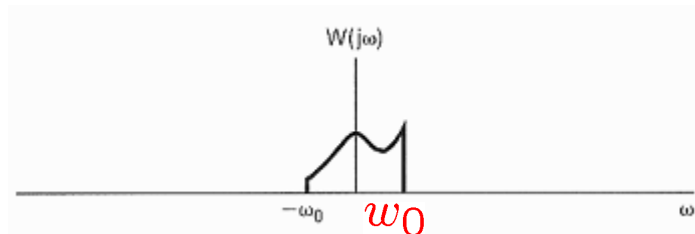
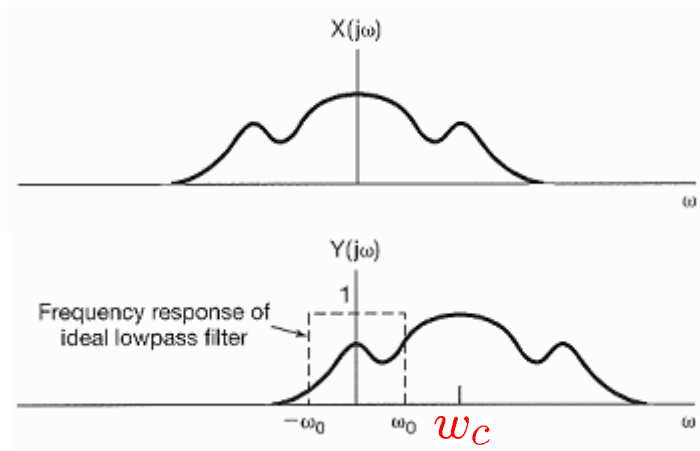
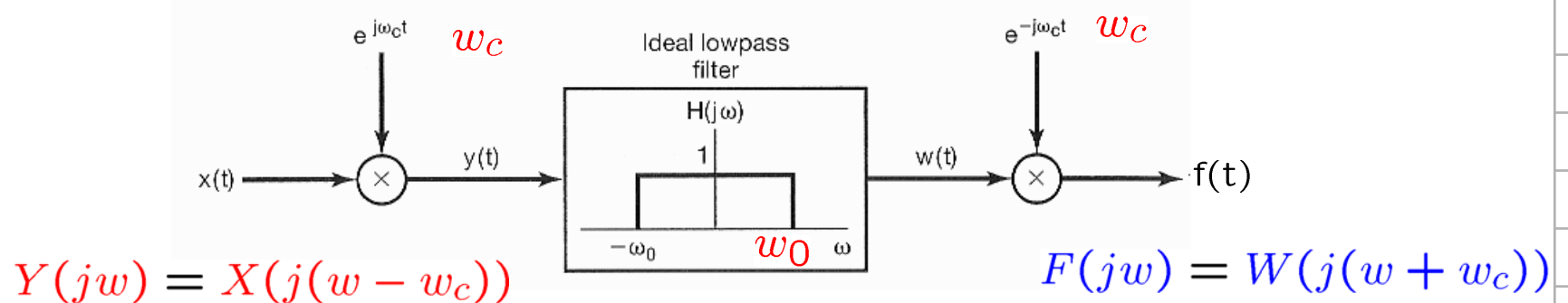


■ Bandpass Filter Using Amplitude Modulation:

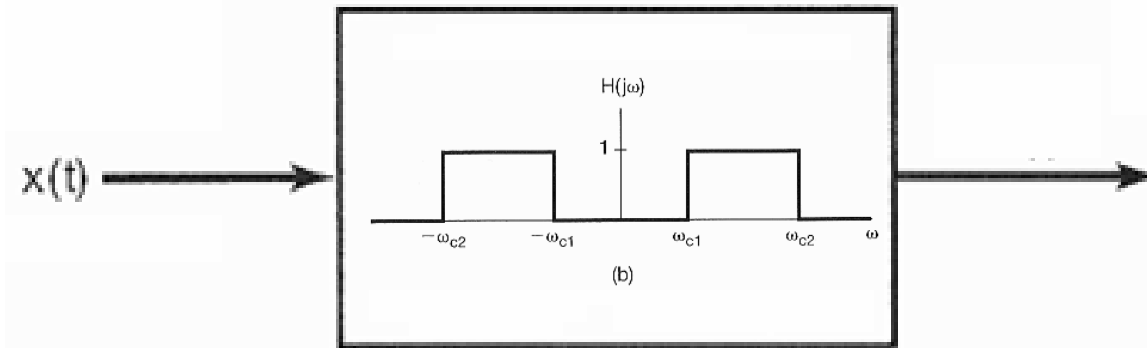


Bandpass Filter Using Amplitude Modulation:

$$e^{j\omega_c t} \xleftrightarrow{\mathcal{F}} 2\pi\delta(\omega - \omega_c)$$



■ Bandpass Filter Using Amplitude Modulation:



- On Page 349-350, Problem 4.46

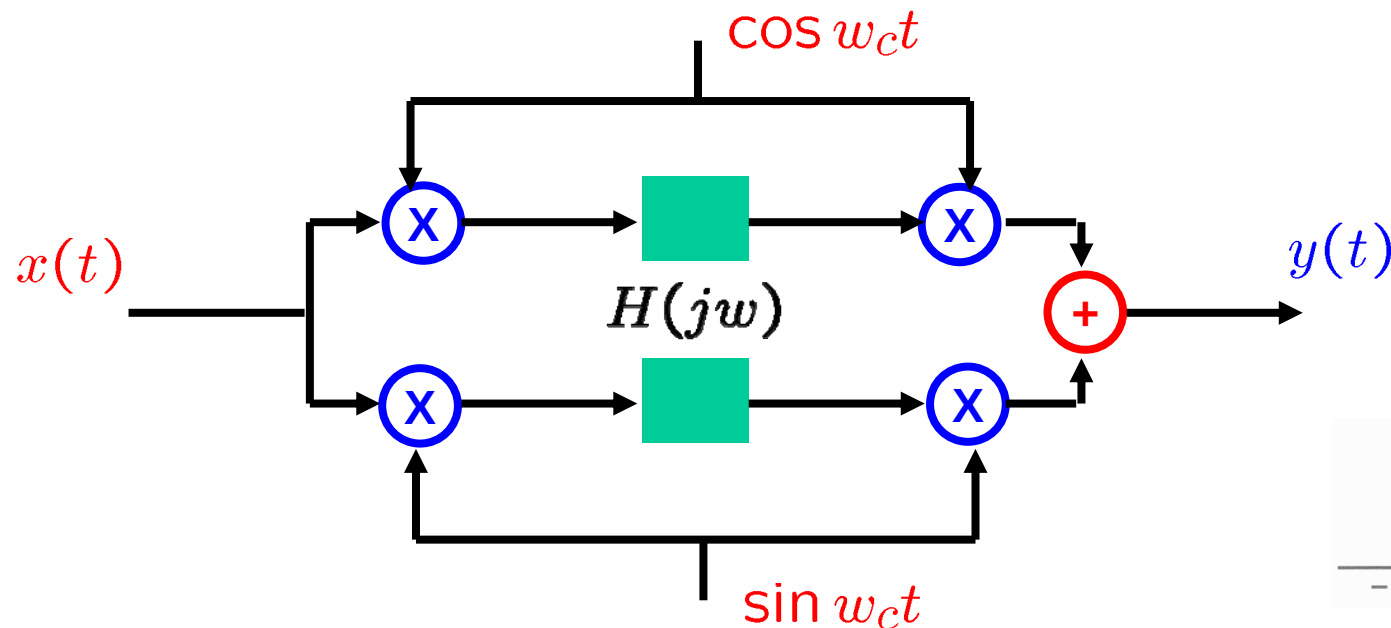


TABLE 4.1 PROPERTIES OF THE FOURIER TRANSFORM

Section	Property	Aperiodic signal	Fourier transform
		$x(t)$ $y(t)$	$X(j\omega)$ $Y(j\omega)$
4.3.1	Linearity	$ax(t) + by(t)$	$aX(j\omega) + bY(j\omega)$
4.3.2	Time Shifting	$x(t - t_0)$	$e^{-j\omega t_0} X(j\omega)$
4.3.6	Frequency Shifting	$e^{j\omega_0 t} x(t)$	$X(j(\omega - \omega_0))$
4.3.3	Conjugation	$x^*(t)$	$X^*(-j\omega)$
4.3.5	Time Reversal	$x(-t)$	$X(-j\omega)$
4.3.5	Time and Frequency Scaling	$x(at)$	$\frac{1}{ a } X\left(\frac{j\omega}{a}\right)$
4.4	Convolution	$x(t) * y(t)$	$X(j\omega)Y(j\omega)$
4.5	Multiplication	$x(t)y(t)$	$\frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\theta)Y(j(\omega - \theta))d\theta$
4.3.4	Differentiation in Time	$\frac{d}{dt} x(t)$	$j\omega X(j\omega)$
4.3.4	Integration	$\int_{-\infty}^t x(t)dt$	$\frac{1}{j\omega} X(j\omega) + \pi X(0)\delta(\omega)$
4.3.6	Differentiation in Frequency	$tx(t)$	$j \frac{d}{d\omega} X(j\omega)$
4.3.3	Conjugate Symmetry for Real Signals	$x(t)$ real	$\begin{cases} X(j\omega) = X^*(-j\omega) \\ \Re\{X(j\omega)\} = \Re\{X(-j\omega)\} \\ \Im\{X(j\omega)\} = -\Im\{X(-j\omega)\} \\ X(j\omega) = X(-j\omega) \\ \angle X(j\omega) = -\angle X(-j\omega) \end{cases}$
4.3.3	Symmetry for Real and Even Signals	$x(t)$ real and even	$X(j\omega)$ real and even
4.3.3	Symmetry for Real and Odd Signals	$x(t)$ real and odd	$X(j\omega)$ purely imaginary and odd
4.3.3	Even-Odd Decomposition for Real Signals	$x_e(t) = \mathcal{E}\{x(t)\}$ [x(t) real] $x_o(t) = \mathcal{O}\{x(t)\}$ [x(t) real]	$\Re\{X(j\omega)\}$ $j\Im\{X(j\omega)\}$
4.3.7	Parseval's Relation for Aperiodic Signals		
	$\int_{-\infty}^{+\infty} x(t) ^2 dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) ^2 d\omega$		

TABLE 4.2 BASIC FOURIER TRANSFORM PAIRS

Signal	Fourier transform	Fourier series coefficients (if periodic)
$\sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$	$2\pi \sum_{k=-\infty}^{+\infty} a_k \delta(\omega - k\omega_0)$	a_k
$e^{j\omega_0 t}$	$2\pi \delta(\omega - \omega_0)$	$a_1 = 1$ $a_k = 0$, otherwise
$\cos \omega_0 t$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$	$a_1 = a_{-1} = \frac{1}{2}$ $a_k = 0$, otherwise
$\sin \omega_0 t$	$\frac{\pi}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$	$a_1 = -a_{-1} = \frac{1}{2j}$ $a_k = 0$, otherwise
$x(t) = 1$	$2\pi \delta(\omega)$	$a_0 = 1$, $a_k = 0$, $k \neq 0$ (this is the Fourier series representation for any choice of $T > 0$)
Periodic square wave		
$x(t) = \begin{cases} 1, & t < T_1 \\ 0, & T_1 < t \leq \frac{T}{2} \end{cases}$ and $x(t + T) = x(t)$	$\sum_{k=-\infty}^{+\infty} \frac{2 \sin k\omega_0 T_1}{k} \delta(\omega - k\omega_0)$	$\frac{\omega_0 T_1}{\pi} \operatorname{sinc}\left(\frac{k\omega_0 T_1}{\pi}\right) = \frac{\sin k\omega_0 T_1}{k\pi}$

and

$$x(t + T) = x(t)$$

$$\sum_{n=-\infty}^{+\infty} \delta(t - nT) \quad \frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta\left(\omega - \frac{2\pi k}{T}\right) \quad a_k = \frac{1}{T} \text{ for all } k$$

$$x(t) \begin{cases} 1, & |t| < T_1 \\ 0, & |t| > T_1 \end{cases} \quad \frac{2 \sin \omega T_1}{\omega} \quad \text{—}$$

$$\frac{\sin Wt}{\pi t} \quad X(j\omega) = \begin{cases} 1, & |\omega| < W \\ 0, & |\omega| > W \end{cases} \quad \text{—}$$

$$\delta(t) \quad 1 \quad \text{—}$$

$$u(t) \quad \frac{1}{j\omega} + \pi \delta(\omega) \quad \text{—}$$

$$\delta(t - t_0) \quad e^{-j\omega t_0} \quad \text{—}$$

$$e^{-at} u(t), \operatorname{Re}\{a\} > 0 \quad \frac{1}{a + j\omega} \quad \text{—}$$

$$te^{-at} u(t), \operatorname{Re}\{a\} > 0 \quad \frac{1}{(a + j\omega)^2} \quad \text{—}$$

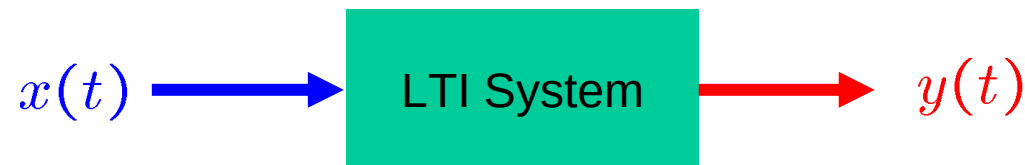
$$\frac{t^{n-1}}{(n-1)!} e^{-at} u(t), \operatorname{Re}\{a\} > 0 \quad \frac{1}{(a + j\omega)^n} \quad \text{—}$$

- Representation of **Aperiodic** Signals:
the Continuous-Time Fourier Transform
- The Fourier Transform for **Periodic** Signals
- **Properties** of the Continuous-Time Fourier Transform
- The **Convolution** Property
- The **Multiplication** Property
- **Systems** Characterized by
Linear Constant-Coefficient Differential Equations

■ A useful class of CT LTI systems:

$$a_N \frac{d^N y(t)}{dt^N} + a_{N-1} \frac{d^{N-1} y(t)}{dt^{N-1}} + \cdots + a_1 \frac{dy(t)}{dt} + a_0 y(t) = b_M \frac{d^M x(t)}{dt^M} + b_{M-1} \frac{d^{M-1} x(t)}{dt^{M-1}} + \cdots + b_1 \frac{dx(t)}{dt} + b_0 x(t)$$

$$\sum_{k=0}^N a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^M b_k \frac{d^k x(t)}{dt^k}$$



$$Y(j\omega) = X(j\omega)H(j\omega) \quad H(j\omega) = \frac{Y(j\omega)}{X(j\omega)}$$

$$\sum_{k=0}^N a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^M b_k \frac{d^k x(t)}{dt^k}$$

$$\sum_{k=0}^N a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^M b_k \frac{d^k x(t)}{dt^k}$$

$$\sum_{k=0}^N a_k (j\omega)^k Y(j\omega) = \sum_{k=0}^M b_k (j\omega)^k X(j\omega)$$

$$Y(j\omega) \left[\sum_{k=0}^N a_k (j\omega)^k \right] = X(j\omega) \left[\sum_{k=0}^M b_k (j\omega)^k \right]$$

$$\Rightarrow H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{\sum_{k=0}^M b_k (j\omega)^k}{\sum_{k=0}^N a_k (j\omega)^k} = \frac{b_M (j\omega)^M + \dots + b_1 (j\omega) + b_0}{a_N (j\omega)^N + \dots + a_1 (j\omega) + a_0}$$

■ Examples 4.24 & 4.25:

$$\frac{dy(t)}{dt} + ay(t) = x(t)$$

$$H = \frac{Y}{X}$$

$$\Rightarrow H(j\omega) = \frac{1}{j\omega + a}$$

$$(j\omega)Y(j\omega) + aY(j\omega) = X(j\omega)$$

$$\Rightarrow h(t) = e^{-at}u(t)$$

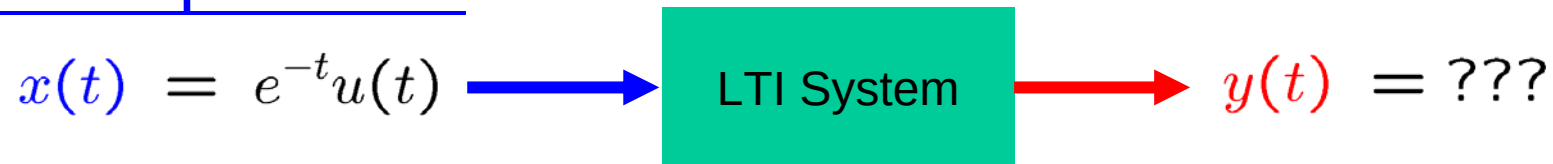
$$\frac{d^2y(t)}{dt^2} + 4\frac{dy(t)}{dt} + 3y(t) = \frac{dx(t)}{dt} + 2x(t)$$

$$\Rightarrow H(j\omega) = \frac{(j\omega) + 2}{(j\omega)^2 + 4(j\omega) + 3} = \frac{(j\omega + 2)}{(j\omega + 1)(j\omega + 3)}$$

$$= \frac{1/2}{j\omega + 1} + \frac{1/2}{j\omega + 3}$$

$$\Rightarrow h(t) = \frac{1}{2}e^{-t}u(t) + \frac{1}{2}e^{-3t}u(t)$$

■ Example 4.26:



$$H(j\omega) = \frac{(j\omega + 2)}{(j\omega + 1)(j\omega + 3)}$$

$$\Rightarrow Y(j\omega) = X(j\omega)H(j\omega)$$

$$= \left[\frac{1}{j\omega + 1} \right] \left[\frac{j\omega + 2}{(j\omega + 1)(j\omega + 3)} \right]$$

$$= \frac{j\omega + 2}{(j\omega + 1)^2(j\omega + 3)}$$

$$= \frac{\frac{1}{4}}{j\omega + 1} + \frac{\frac{1}{2}}{(j\omega + 1)^2} - \frac{\frac{1}{4}}{j\omega + 3}$$

$$\Rightarrow y(t) = \left[\frac{1}{4} e^{-t} + \frac{1}{2} t e^{-t} - \frac{1}{4} e^{-3t} \right] u(t)$$

- Representation of Aperiodic Signals: the CT FT
- The FT for Periodic Signals
- Properties of the CT FT
 - Linearity
 - Conjugation
 - Convolution
 - Differentiation in Time
 - Conjugate Symmetry for Real Signals
 - Symmetry for Real and Even Signals & for Real and Odd Signals
 - Even-Odd Decomposition for Real Signals
 - Parseval's Relation for Aperiodic Signals
 - Time Shifting
 - Time Reversal
 - Multiplication
 - Integration
 - Frequency Shifting
 - Time and Frequency Scaling
 - Differentiation in Frequency
- The Convolution Property
- The Multiplication Property
- Systems Characterized by Linear Constant-Coefficient Differential Equations

- Why to study FT
 - In order to analyze or represent aperiodic signals
- How to develop FT
 - From FS and let $T \rightarrow \text{infinity}$
- Do periodic signals have FT
 - Yes, their FT is function of isolated impulses
- Why to know the properties of FT
 - Avoid using the fundamental formulas of FT to compute the FT
- What the duality of FT and why
 - FT and IFT have almost identical integration formulas
- Why to know the convolution property
 - To analyze system response and/or design proper circuits
 - To simplify computation
- Why to know the multiplication property
 - For signal modulation with different-frequency carriers
 - To simplify computation

$$a_k = \frac{1}{T} X(j\omega) \Big|_{\omega = k\omega_0}$$

$$X(j\omega) = \sum_{k=-\infty}^{+\infty} 2\pi a_k \delta(\omega - k\omega_0)$$

$$= \sum_{k=-\infty}^{+\infty} 2\pi \frac{1}{T} X(jk\omega_0) \delta(\omega - k\omega_0)$$

$$\omega = m\omega_0$$

$$X(jm\omega_0) = \sum_{k=-\infty}^{+\infty} 2\pi \frac{1}{T} X(jk\omega_0) \delta(m\omega_0 - k\omega_0)$$

$$= 2\pi \frac{1}{T} X(jm\omega_0)$$

$$\Rightarrow 2\pi = T$$

$$a_k = \frac{1}{T} X_a(jw) \Big|_{w=kw_0}$$

$$X_p(jw) = \sum_{k=-\infty}^{+\infty} 2\pi a_k \delta(w - kw_0)$$

$$= \sum_{k=-\infty}^{+\infty} 2\pi \frac{1}{T} X_a(jkw_0) \delta(w - kw_0)$$

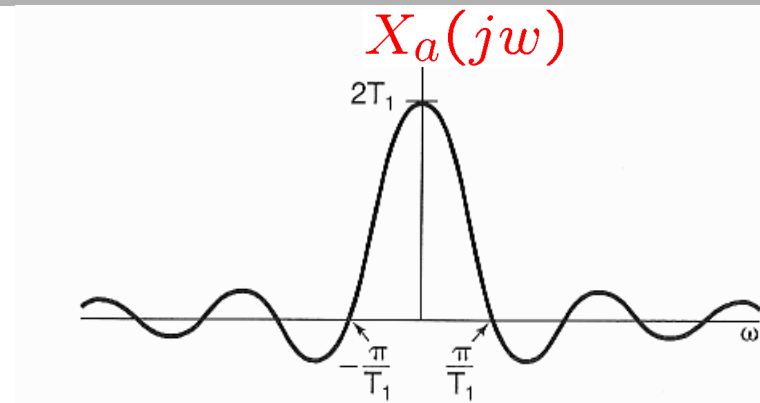
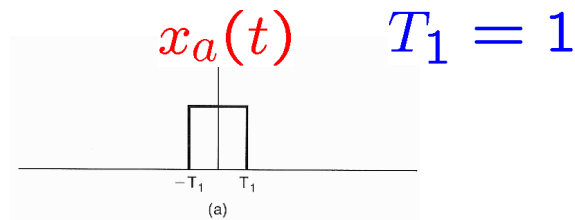
$$w = mw_0$$

$$X_p(jmw_0) = \sum_{k=-\infty}^{+\infty} 2\pi \frac{1}{T} X_a(jkw_0) \delta(mw_0 - kw_0)$$

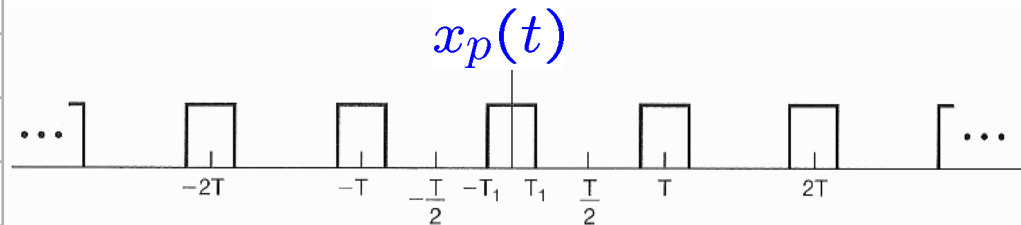
$$= 2\pi \frac{1}{T} X_a(jmw_0)$$

X(jw) of Aperiodic Signals and a_k of Periodic Signals

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$T = 4$ $w_0 = 2\pi/4 = \pi/2$ $X_a(jw) = 2 \frac{\sin(wT_1)}{w} = 2 \frac{\sin(w)}{w}$



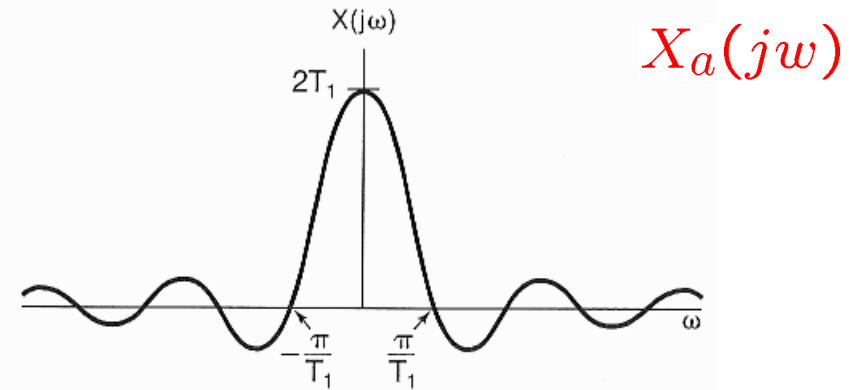
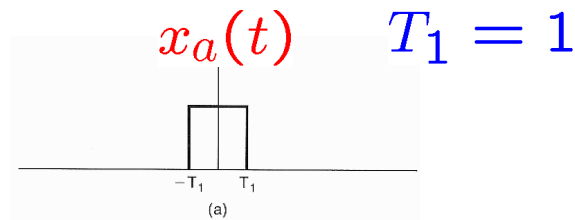
$$\Rightarrow a_k = \frac{1}{T} X_a(jw) \Big|_{w=kw_0} = \frac{\sin(k\pi/2)}{\pi k}$$

$$\Rightarrow X_p(jw) = \sum_{k=-\infty}^{+\infty} 2\pi a_k \delta(w - kw_0) = \sum_{k=-\infty}^{+\infty} \frac{2 \sin(k\pi/2)}{k} \delta(w - kw_0)$$

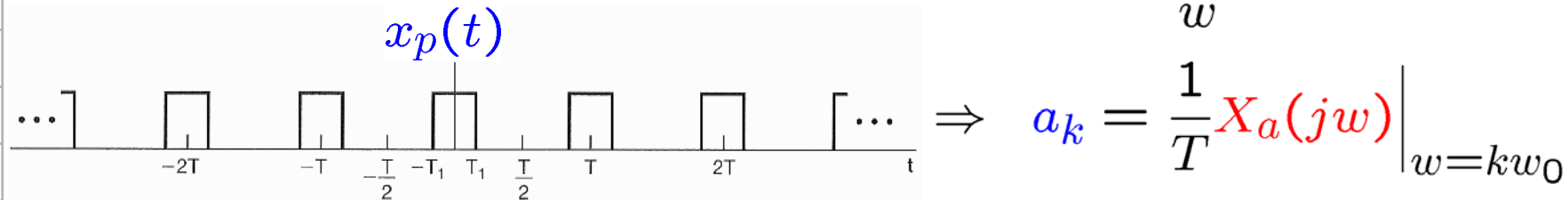
$$\Rightarrow X_p(jmw_0) = \frac{2 \sin(m\pi/2)}{m}$$

X(jw) of Aperiodic Signals and a_k of Periodic Signals

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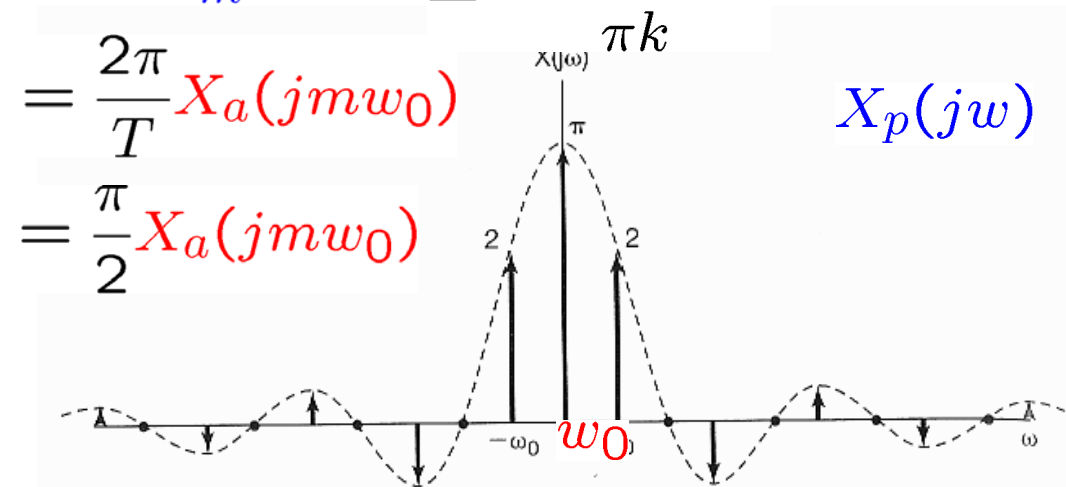


$T = 4$ $\omega_0 = 2\pi/4 = \pi/2$ $X_a(j\omega) = 2 \frac{\sin(\omega T_1)}{\omega} = 2 \frac{\sin(\omega)}{\omega}$



$\Rightarrow X_p(jm\omega_0) = \frac{2 \sin(m\pi/2)}{m} = 2\pi a_m = \frac{\sin(k\pi/2)}{\pi k}$

m	0	1
a_m	1/2	1/\pi
$2\pi a_m$	π	2
$X_p(jm\omega_0)$	π	2
$X_a(jm\omega_0)$	2	4/\pi



Signals & Systems [\(Chap 1\)](#)

LTI & Convolution [\(Chap 2\)](#)

Bounded/Convergent

Periodic

FS

[\(Chap 3\)](#)

– CT
– DT

Aperiodic

FT

– CT [\(Chap 4\)](#)
– DT [\(Chap 5\)](#)

Unbounded/Non-convergent

LT

– CT [\(Chap 9\)](#)

zT

– DT [\(Chap 10\)](#)

Time-Frequency [\(Chap 6\)](#)

Communication [\(Chap 8\)](#)

CT-DT [\(Chap 7\)](#)

Control [\(Chap 11\)](#)